Investigating Electron Transfer across Single-Molecule Junctions

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Abstract

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Electron transfer processes are investigated through conductance measurements of single molecules. Measurements are performed on metal-molecule-metal junctions using a modified scanning tunneling microscope technique. Through a series of experimental measurements, and accompanying theoretical models, the influence of the molecule on the measured current is explored. These explorations are presented in five separate chapters. In chapter two, the molecular orbitals of sp-hybridized carbon chains are discussed in detail. It is demonstrated that the molecular orbitals can assume an intriguing helical shape. In chapter three, the length-dependent conductance of sp^2-hybridized carbon chains is investigated. Experiment and theory demonstrate that the conductance of odd-numbered chains is nearly uniform with length. In chapter four, a new theoretical scheme to calculate quantum interference is developed. Using this scheme, it is demonstrated that quantum interference yields the decay in conductance with length for molecular wires. In chapter five, current-voltage measurements of redox-active molecular clusters are shown to agree with a hopping transport model. In chapter six, a novel experimental setup is presented that can be used to investigate photoconductivity in single-molecule junctions. This thesis provides a broad, yet rigorous, survey of electron transfer processes in single-molecule junctions.

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My journey through science has been long and winding. I appreciate all who supported me during these formative graduate school years as I transitioned from student to scientist.

Dedication

To Mommy, Daddy, and Akka

Chapter 1: Introduction

1.1 Preface

This chapter begins with a brief overview of electron transfer reactions, followed by an introduction to the STM break-junction technique, and ends with a detailed section on the theory of electron transfer across single-molecule junctions. The theory section has been included primarily to memorialize my own understanding of the theory, a theory which has both deeply confused and immensely captivated me during my PhD. I would like to acknowledge invaluable discussions I have had with E-Dean Fung, Daniel Hernangómez-Pérez, Ferdinand Evers, and Latha Venkataraman in preparation of this chapter.

1.2 Electron transfer reactions in chemistry

Electron transfer (ET) reactions are ubiquitous in chemical systems.[1-3] In an ET reaction (Figure 1.1), an electron from a donor (D) is transferred to an acceptor (A) via a bridge (B). The bridge may be a covalent linkage between the donor and acceptor (i.e., "inner sphere" ET),[4] or the bridge may simply represent the solvent separating the donor and acceptor (i.e., "outer sphere" ET).[5] ET reactions can be partitioned into three general categories based on the nature of the donor and acceptor (Figure 1.1).[6] The three categories are: (i) homogenous ET between two molecules,[7-8] (ii) heterogeneous ET between a molecule and an electrode,[9-10] and (iii) homogeneous ET between two electrodes.[11-12] The third category is often regarded as electron transport as opposed to electron transfer since electrons continually flow between the electrodes.[13-14]

From a theoretical standpoint, the distinction between molecule and electrode is two-fold. First, a molecule possesses a discrete spectrum of energy levels, while an electrode possesses a continuous spectrum of energy levels. Second, molecules experience significant distortions in

nuclear structure as a result of ET, whereas electrodes do not.[15] These distinctions yield different regimes of ET theory for each category in Figure 1.1. For ET between two electrodes (iii), the rate of ET depends predominantly on the properties of the bridge.[16]

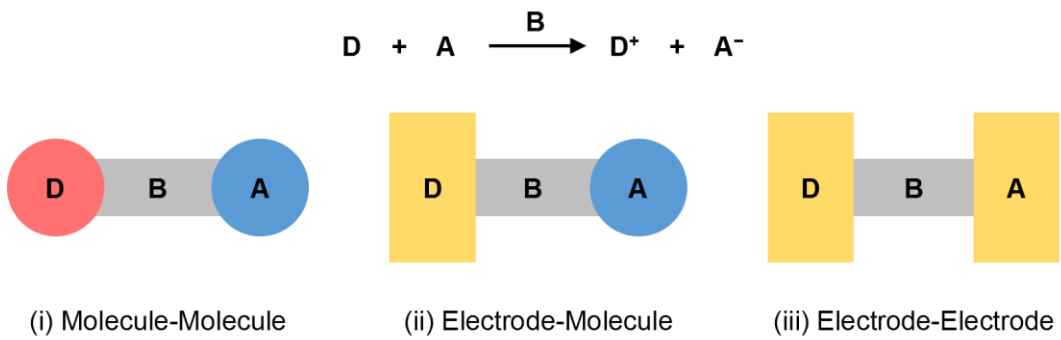

Figure 1.1. Overview of electron transfer (ET) reactions in chemistry.[2] The reactants are a donor (D) and an acceptor (A). The D and A pair may be (i) a molecule and a molecule, (ii) an electrode and a molecule, or (iii) an electrode and an electrode. ET from D to A is facilitated by a bridge (B), and the reaction products are D+ and A−.

The theory of ET has been developed in tandem with increasingly precise experimental techniques. Spectroscopic techniques have enabled measurements of ET with sub-picosecond temporal resolution,[17-18] while scanning probe techniques have enabled measurements of ET with atomic-scale spatial resolution.[19-20] Together, these experimental techniques have provided an intricate picture of ET, through both time and space, for a host of chemical systems. In this thesis, I will discuss the experiment and theory of ET across single-molecule junctions, where a single molecule serves as the bridge between two metal electrodes.[21-22]

1.3 Scanning tunneling microscope break-junction technique

Single-molecule junctions are formed using a modified scanning tunneling microscope (STM) technique.[21, 23] A mechanically cut gold wire (250 μm diameter) and a gold-coated steel substrate (0.1–0.2 μm film thickness) serve as the two electrodes. A fixed voltage (V) is applied between the electrodes, and the current (I) is measured (typically 40 kHz acquisition rate). The conductance of the circuit, G = I/V, is typically presented in units of $G_0 = 2e^2/h$.

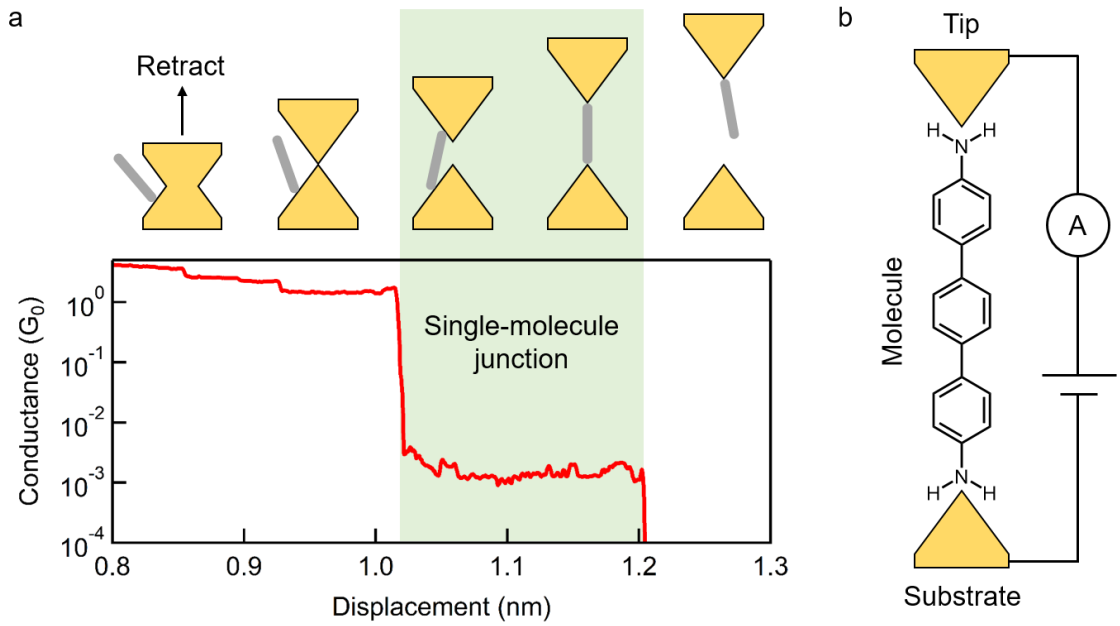

Figure 1.2. a) Conductance trace for a single break-junction measurement. The conductance plateau indicative of a single-molecule junction is highlighted in green. b) Schematic of a single-molecule junction. Linker groups, e.g., amines, enable the molecule to bridge the electrodes.

Measurements are performed by depositing a solution of the target molecule on the substrate (~ 0.1 mM). Using a piezoelectric motor, the gold tip is brought into mechanical contact with the substrate until the conductance exceeds a threshold conductance of 5 G_0. The gold tip is then retracted from the substrate at a speed of 20 nm/s and conductance versus tip displacement is recorded (Figure 1.2a). As the tip is retracted, the conductance decreases in discrete steps of G_0, in accordance with the number of atoms in the contact. When the conductance equals 1 G_0, the contact comprises one gold atom.[24] With further tip retraction, the gold contact breaks and the electrodes snapback, yielding a ~ 0.5 nm gap that can be bridged by a single molecule from the surrounding solution.[25] The molecule can bind to the gold electrodes provided that the molecule contains aurophilic linker groups (e.g., thiols, thioethers, and amines). The measured conductance (< 1 G_0) represents the conductance of a single-molecule junction (Figure 1.2b). The single-molecule junction is maintained for some amount of additional tip retraction until the junction

ruptures, yielding a plateau in the conductance trace. This modified STM technique, referred to as the STM break-junction technique, provides a facile method for direct measurement of ET across a single-molecule bridge.[26]

1.4 Theory of electron transfer across molecular junctions

The ultimate aim of ET theory for molecular junctions is to calculate the function $I(V)$, which describes the relationship between the measured current I and the applied voltage V. Interestingly, the current does not vary linearly with voltage, per Ohms law: $I(V) = GV$, with fixed conductance G. Rather, $I(V)$ can be highly nonlinear. The proper description of $I(V)$ for a molecular junction derives from the more general topic of quantum scattering theory.[27] An electron incident on the molecular junction from one electrode can scatter off the junction and be transferred, with some probability, to the other electrode. The scattering process is often assumed to be elastic, meaning that the energy of the electron is the same before and after the scattering event. Moreover, the scattering process is coherent, meaning that the incoming and outgoing electron waves have a well-defined phase relation—i.e., the wave function of the scattering state is delocalized throughout the entire molecular junction. It is the coherence of the electron wave that yields a nonlinear $I(V)$ by way of resonance and interference effects.[28-29]

1.4.1 Landauer formalism

The theoretical approach used to model molecular junctions, originally conceived by Landauer,[30] is presented in Figure 1.3. The molecular junction is connected to a reservoir of electrons on the left (L) and the right (R). The electrons in the reservoirs obey Fermi-Dirac statistics. The probability of an electron occupying a single-particle state i with energy E_i is given by $f(E_i) = 1/\left(1 + e^{(E_i-\mu)/k_BT}\right)$, where k_B is Boltzmann's constant, T is the temperature, and μ is the chemical potential of the reservoir. If the left and right reservoirs are in chemical and thermal

equilibrium ($\mu_L = \mu_R$ and $T_L = T_R$), electrons can flow between the reservoirs, but there is no net electrical current. Biasing the junction by applying a voltage V, such that $\mu_L - \mu_R = eV$, results in a net current as the system attempts to reach chemical equilibrium ($e = -|e|$ is the electric charge).[31] The net current can be expressed as an integral over energy (Eq. 1.1), governed by a transmission function $T(E)$ (Figure 1.3).

$$I = \frac{2e}{h} \int dE \big(f_L(E) - f_R(E)\big) T(E) \tag{1.1}$$

Eq. 1.1 assumes that electron transmission from left to right is the same as from right to left ($T_{L\to R}(E) = T_{R\to L}(E)$). Additionally, Eq. 1.1 assumes transmission is identical for both spin up and spin down electrons ($T_\uparrow(E) + T_\downarrow(E) = 2T(E)$), yielding a factor of two in the prefactor.

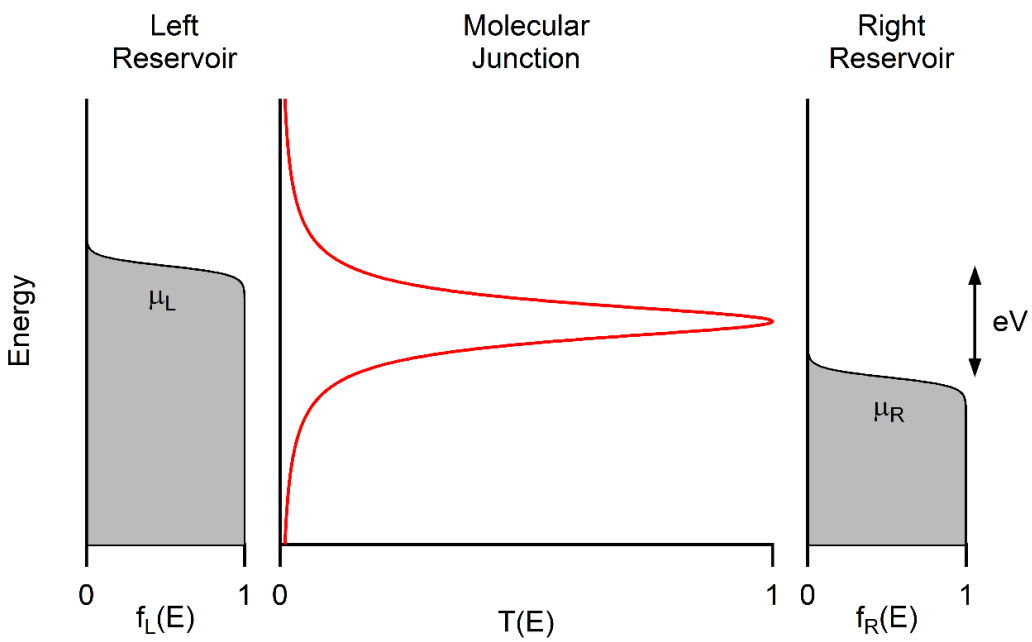

Figure 1.3. Schematic of the Landauer formalism for calculating the current through a molecular junction. The molecule and the electrodes are described by a transmission function $T(E)$. Electrons are supplied by reservoirs that are held at different chemical potentials (μ_L and μ_R).

All details of the molecular junction are placed into the transmission function, and the Landauer formalism transforms the calculation of $I(V)$ into a calculation of $T(E)$. It is often

assumed that the transmission function is independent of applied voltage: $T(E,V) \approx T^0(E)$, where $T^0(E)$ denotes the zero-bias transmission function. Hence, the voltage only modifies the chemical potentials of the reservoirs, which are often symmetrically displaced from the Fermi Energy (E_F): $\mu_L = E_F + eV/2$ and $\mu_R = E_F - eV/2$. However, it should be noted that recent experiments suggest that, for measurements in ionic solutions, the transmission function can exhibit a strong bias dependence. Due to the presence of ions, it is believed that the applied voltage serves both as a source-drain bias and as a gate voltage. In that case, the voltage-dependent transmission function has been modeled as $T(E,V) = T^0(E + \alpha eV)$, where α describes the gating strength.[32-33]

1.4.2 Conductance quantization

A key result of the Landauer formalism is that the conductance of a one-dimensional conductor is quantized.[34-35] A simple heuristic argument can explain conductance quantization using the time-energy uncertainty principle. Consider applying an infinitesimal voltage difference dV across a molecular junction, which opens up a bias window of $\mu_L - \mu_R = edV$ around the Fermi energy E_F. The current is maximized if the molecule possesses an energy eigenstate $|m\rangle$ with energy E_m that is within the bias window (i.e., $E_m = E_F$). The current produced by g electrons, with energy E_F, incident on the junction depends both on the density of states $\rho(E_F)$ of the molecule and the lifetime τ of the electrons on the molecule (Eq. 1.2). The density of states at E_F can be expressed as $\rho(E_F) = 1/\Delta E$, where the energy level of the molecule is smeared out over some energy width ΔE.

$$I(dV) = \underbrace{ge}_{\substack{electric\\charge}} \times \underbrace{(1/\tau)}_{\substack{inverse\ of\\lifetime}} \times \underbrace{(1/\Delta E)}_{\substack{density\\of\ states}} \times \underbrace{(\mu_L - \mu_R)}_{\substack{bias\\window}} \leq g(e^2/h)dV \qquad (1.2)$$

If the molecule is not connected to the electrodes, then the energy eigenstate of the molecule has definite energy ($\Delta E \to 0$) and infinite lifetime ($\tau \to \infty$). No current would be

measured since the electron could not enter or exit the molecule. To pass a current through the molecule, the lifetime must be reduced by connecting, or "coupling", the molecule to the electrodes. Naively, one might expect that the current could be increased arbitrarily by reducing the lifetime. However, the lifetime cannot be reduced independently. Coupling the molecule to the electrodes produces a level broadening (ΔE) commensurate with the reduced lifetime (τ). This is codified by the time-energy uncertainty principle: $\tau \Delta E \geq h$ (based on the definition of τ and ΔE for this heuristic). Plugging in $\tau \Delta E \geq h$ into Eq. 1.2, and using $I(dV) = GdV$, reveals a maximum conductance of $G = g(e^2/h)$.

The factor g represents the number of degenerate states with energy E_F, and conductance quantization arises from the fact that g can be any positive integer.[36-37] For spin degenerate systems, g must be even since each electronic state can host two electrons (spin up/down). Hence, it is useful to define the conductance quantum as $G_0 = 2e^2/h$. In addition to spin degeneracy, there can be electronic degeneracy. Electronic degeneracy gives rise to conductance channels each with a maximum conductance of G_0 (assuming spin degeneracy). For example, if there are N conductance channels, then $g = 2N$ and the maximum conductance is NG_0.

More explicitly, the conductance for channel n can be expressed as $G = G_0 \mathcal{T}_n(E_F)$, where $0 < \mathcal{T}_n < 1$ is the transmission probability.[14] Similarly, the conductance through all N channels can be expressed by Eq. 1.3, where the transmission function $T(E)$ is the sum of the transmission probabilities for all the channels: $T(E) = \sum_{n=1}^{N} \mathcal{T}_n(E)$. For a molecular junction, current flows predominantly through a single channel; hence, $T(E) < 1$ and the conductance is less than 1 G_0.

$$G = G_0 T(E_F) \tag{1.3}$$

Eq. 1.3 establishes the core dictum of the Landauer formalism that "conductance is transmission".[38] The equation for the current (Eq. 1.1) is simply a generalization of this dictum.

1.5 Discussion of the transmission function

To calculate the current using the Landauer formalism, the transmission function $T(E)$ must be calculated. In the following sections, a primer on the theoretical principles that underpin $T(E)$ is presented.[14, 39] The molecular junction, comprising the molecule and the left and right electrodes, is described by a Hamiltonian $\hat{H}$. The transmission function could be readily calculated if the energy eigenstates $|n\rangle$ of $\hat{H}$ were known, where $\hat{H}|n\rangle = E_n|n\rangle$. However, in general, the eigenstates of $\hat{H}$ are hard to compute. A more effective approach is to view $\hat{H}$ as a "perturbed" Hamiltonian (Eq. 1.4), where the Hamiltonian $\hat{V}$ perturbs the "unperturbed" Hamiltonian $\hat{H}_0$ (with known eigenstates). $T(E)$ can be expressed in terms of $\hat{V}$ and the eigenstates of $\hat{H}_0$.

$$\hat{H} = \hat{H}_0 + \hat{V} \tag{1.4}$$

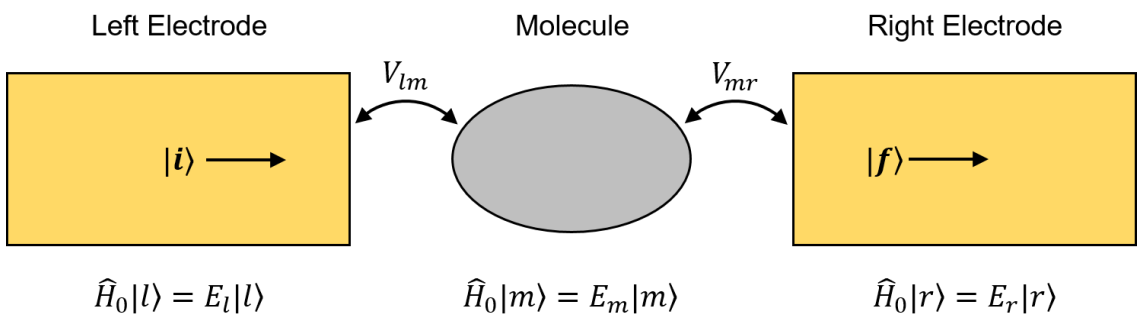

Figure 1.4. Schematic of the Hamiltonian describing a molecular junction ($\hat{H} = \hat{H}_0 + \hat{V}$). Current can be modeled as a set of transition rates from an initial state $|i\rangle$ in the left electrode to a final state $|f\rangle$ in the right electrode. States $|i\rangle$ and $|f\rangle$ denote specific states from the collection of $|l\rangle$ and $|r\rangle$, respectively. Matrices are denoted as $\hat{M}$ with matrix elements $M_{ij} = \langle i|\hat{M}|j\rangle$.

For a molecular junction, $\hat{H}_0$ describes the molecule and the left and right electrodes uncoupled from each other (Figure 1.4). The eigenstates of $\hat{H}_0$ fall into three categories: states $|l\rangle$ that are localized on the left electrode (with energy E_l), states $|m\rangle$ that are localized on the molecule (with energy E_m), and states $|r\rangle$ that are localized on the right electrode (with energy E_r). The Hamiltonian $\hat{V}$ describes the coupling between the molecule and the electrodes (Figure

1.4). The only nonzero matrix elements of $\hat{V}$ are between states on the molecule ($|m\rangle$) and states on either electrode ($|l\rangle$ or $|r\rangle$).

1.5.1 Fermi's golden rules

It is convenient to use *Fermi's golden rule 2* (Eq. 1.5) and *Fermi's golden rule 1* (Eq. 1.6) as a starting point for a discussion of $T(E)$.[40-42] These golden rules describe the transition rate between two energy eigenstates of Hamiltonian $\hat{H}_0$ due the perturbation $\hat{V}$. The perturbation enables an initial eigenstate $|i\rangle$ that is localized on the left electrode (i.e., some state $|l\rangle$) to transition to a final eigenstate $|f\rangle$ that is localized on the right electrode (i.e., some state $|r\rangle$). The transition rate (with units: 1/time) is denoted as $w_{i\to f}$. Often the exact transition rate $w_{i\to f}$ is inaccessible, so Fermi's golden rules approximate the transition rate to the lowest order of $\hat{V}$. Fermi's golden rule 2 (Eq. 1.5) describes the first-order transition rate ($w_{i\to f}^{(1)}$), and Fermi's golden rule 1 (Eq. 1.6) describes the second-order transition rate ($w_{i\to f}^{(2)}$). In the final section, a generalization of these golden rules is presented to describe the exact transition rate.

Fermi's golden rule 2 (Eq. 1.5) describes a direct transition from $|i\rangle$ to $|f\rangle$. The direct transition is dictated by the matrix element $V_{fi} = \langle f|V|i\rangle$. For this reason, matrix elements of V are often regarded as "hopping" terms. The Dirac delta function $\delta(x)$ imposes energy conservation on the transition ($E_f = E_i$). Per the theoretical setup in Figure 1.4, the direct coupling between the two electrodes is zero ($V_{fi} = 0$), so the first-order transition rate is zero ($w_{i\to f}^{(1)} = 0$).

$$w_{i\to f}^{(1)} = \frac{2\pi}{\hbar}\left|V_{fi}\right|^2 \delta\left(E_f - E_i\right) \tag{1.5}$$

Since $w_{i\to f}^{(1)} = 0$, the leading order of $w_{i\to f}$ falls to the second order transition rate $w_{i\to f}^{(2)}$. The second order transition rate is given by Fermi's golden rule 1 (Eq. 1.6). Golden rule 1 describes a sum of indirect transitions from $|i\rangle$ to $|f\rangle$ via an intermediate state $|m\rangle$. In general, the summation

(Eq. 1.6) runs over all eigenstates of $\widehat{H}_0$ (i.e., $|l\rangle$, $|m\rangle$, and $|r\rangle$). However, for a molecular junction, only the eigenstates that reside on the molecule (i.e., $|m\rangle$) couple to both electrodes (i.e., V_{ml} and V_{mr} are nonzero). Thus, only the eigenstates of the molecule (i.e., the molecular orbitals, MOs) participate in the indirect transition. In this way, the MOs of the molecular bridge mediate ET between the electrodes. In chemistry parlance, bridge-mediated ET is referred to as superexchange.

$$w_{i\to f}^{(2)} = \frac{2\pi}{\hbar}\left|\sum_m \frac{V_{fm}V_{mi}}{E_i - E_m}\right|^2 \delta\left(E_f - E_i\right) \tag{1.6}$$

Interestingly, the Dirac delta function in golden rule 1 (Eq. 1.6) imposes energy conservation between the initial and final states ($E_i = E_f$), but golden rule 1 does not require energy conservation for the intermediate state (E_m). For this reason, the non-conserving transitions $|i\rangle \to |m\rangle$ and $|m\rangle \to |f\rangle$ can be viewed as "virtual" transitions.[43] These virtual transitions are more favorable if the intermediate state is close in energy to the initial/final states, since energy conservation is less severely violated. Accordingly, the denominator in Eq. 1.6 indicates that the rate of ET increases substantially if one of the energy levels (E_m) approaches E_i, a phenomenon referred to as *resonance* (ignore for the moment the singularity at $E_i = E_m$).

Another intriguing phenomenon that appears in golden rule 1 (Eq. 1.6) is *quantum interference (QI)*. The terms in the summation can have differing signs (+/-) and therefore can add together (constructive QI) or cancel each other out (destructive QI). For example, it is possible to have complete destructive QI where the sum evaluates to zero even if all of the individual terms are nonzero. The sign of each term in the summation depends on both the sign of the denominator ($E_i - E_m$) and the sign of the numerator ($V_{fm}V_{mi}$). The sign of the numerator can be simplified if the electrodes each couple to only a single atomic site on the molecule. Suppose that the left electrode only couples to site 1 and the right electrode only couples to site N. The coupling terms

in the numerator can then be expressed as,

$$V_{mi} = V_{i1} C_1^{m*} \tag{1.7}$$

$$V_{fm} = V_{fN} C_N^m \tag{1.8}$$

where C_1^m and C_N^m are coefficients of the mth MO on site 1 and site N, respectively (the * denotes complex conjugate). Using Eq. 1.7 and Eq. 1.8, the terms V_{i1} and V_{fN} can be factored from the summation of Eq. 1.6, simplifying golden rule 1 to Eq. 1.9. Eq. 1.9 demonstrates that the sign of the numerator depends on the product of the MO coefficients on the sites that couple to the electrodes.

$$w_{i\to f}^{(2)} = \frac{2\pi}{\hbar} \left|V_{fN}\right|^2 |V_{i1}|^2 \left|\sum_m \frac{C_N^m C_1^{m*}}{E_i - E_m}\right|^2 \delta\left(E_f - E_i\right) \tag{1.9}$$

The product of $(C_N^m)(C_1^{m*})$ is positive if the MO coefficients have the same phase (e.g., $(+)(+)$ or $(-)(-)$) and is negative if they have opposite phase (e.g., $(+)(-)$ or $(-)(+)$). Therefore, the symmetry of the MOs influences QI.[44] All of the basic qualitative phenomena that underpin the transmission function ($T(E)$) can be understood from Fermi's golden rule 1 (Eq. 1.6).

1.5.2 Level broadening

To calculate $T(E)$ quantitatively, an expression for the exact transition rate $w_{i\to f}$ is required. Fermi's golden rule 1 (Eq. 1.6) is almost correct, but it needs to be modified. To simplify the discussion, the summation over m will be dropped, and it will be assumed that the molecule only has a single energy level (E_m). When $E_i = E_m$, Fermi's golden rule 1 breaks down because state $|m\rangle$ no longer has a definite energy (E_m) as a result of the perturbation $\hat{V}$. For the second-order transition rate, $|m\rangle$ is approximated as having a definite energy E_m, but for the exact transition rate such an approximation cannot be made. Coupling the molecule to the electrodes results in a broadening of the energy level. The effect of level broadening can be inserted into

golden rule 1 (Eq. 1.6) by making the substitution: $E_m \rightarrow E_m - i\,\Gamma_m/2$. With this substitution, the transition rate through a single level takes form of Eq. 1.10.

$$w_{i\rightarrow f} = \frac{2\pi}{\hbar}\left|\frac{V_{fm}V_{mi}}{E_i - (E_m - i\,\Gamma_m/2)}\right|^2 \delta\left(E_f - E_i\right) \tag{1.10}$$

Eq. 1.10 no longer has a singularity at $E_i = E_m$. The use of a complex energy term ($i\,\Gamma_m/2$) is a mathematical trick that makes the most sense in the time domain.[14] The time evolution of an electron initially prepared in state $|m\rangle$, which normally evolves as $|\psi(t)\rangle = e^{-iE_m t/\hbar}|m\rangle$, evolves as $|\psi(t)\rangle = e^{-i(E_m - i\Gamma_m/2)t/\hbar}|m\rangle$ if the substitution $E_m \rightarrow E_m - i\,\Gamma_m/2$ is made. The probability of finding the electron in state $|m\rangle$ sometime t later is given by Eq. 1.11. The probability decays exponentially, with lifetime $\tau = \hbar/\Gamma_m$. Hence, the substitution $E_m \rightarrow E_m - i\,\Gamma_m/2$ describes the concept that the electron can leak into the electrodes due to the coupling $\hat{V}$. This key feature is missing from golden rule 1 (Eq. 1.6).

$$|\langle m|\psi(t)\rangle|^2 = \left|\langle m|e^{-i(E_m - i\Gamma_m/2)t/\hbar}|m\rangle\right|^2 = e^{-\Gamma_m t/\hbar} \tag{1.11}$$

Since the probability of finding the electron on the molecule decays exponentially with time (Eq. 1.11), the leakage can be viewed as a first-order kinetic reaction with rate constant $w_m = \Gamma_m/\hbar$. It is useful to define separate leakage rates for the left ($w_{m\rightarrow L} = \Gamma_m^L/\hbar$) and right ($w_{m\rightarrow L} = \Gamma_m^R/\hbar$) electrodes, where the total leakage rate is simply the sum: $\Gamma_m/\hbar = (\Gamma_m^L + \Gamma_m^R)/\hbar$. The rates $\Gamma_m^L/\hbar$ and $\Gamma_m^R/\hbar$ can be viewed as direct transitions from the molecule to the electrode described by Fermi's golden rule 2 (Eq. 1.5). Since the electron on the molecule does not have a definite energy, the energy of the initial state is left unspecified (E), and $\Gamma_m^L(E)$ and $\Gamma_m^R(E)$ are expressed as functions of E. Since the electron can leak into any electrode state with energy E, the total leakage rate is a sum over all final states in the electrode. Applying golden rule 2 (Eq. 1.5), $\Gamma_m^L(E)$ and $\Gamma_m^R(E)$ are given by Eq. 1.12 and Eq. 1.13, respectively.

$$\Gamma_m^L(E) = 2\pi \sum_l |V_{ml}|^2 \delta(E_l - E) \approx 2\pi \overline{|V_{ml}|^2} \rho_L(E) \tag{1.12}$$

$$\Gamma_m^R(E) = 2\pi \sum_r |V_{mr}|^2 \delta(E_r - E) \approx 2\pi \overline{|V_{mr}|^2} \rho_R(E) \tag{1.13}$$

In the second part of Eq. 1.12 and Eq. 1.13, the average values $\overline{|V_{ml}|^2}$ and $\overline{|V_{mr}|^2}$ (for states with energy E) have been pulled out of the summation. In so doing, $\Gamma_m^L(E)$ and $\Gamma_m^R(E)$ can be expressed in terms of the electrode density of states, $\rho_L(E) = \sum_l \delta(E - E_l)$ and $\rho_R(E) = \sum_r \delta(E - E_r)$, respectively. While $\Gamma_m^L(E)/\hbar$ and $\Gamma_m^R(E)/\hbar$ were presented as rates of leakage off of the molecule, these terms also provide the hopping rates onto the molecule for an electron with energy E.

For simplicity, it is often assumed that the density of states is constant near the Fermi energy (i.e., $\rho_L(E) = \rho_L(E_F)$ and $\rho_R(E) = \rho_R(E_F)$), which is a good assumption for gold electrodes. This so-called "wide-band limit" leaves Γ_m^L and Γ_m^R energy independent. Furthermore, if it is assumed that the electrodes only couple to sites 1 and N on the molecule, Eq. 1.7 and Eq. 1.8 can be applied to arrive at simplified expressions for Γ_m^L and Γ_m^R (Eq. 1.14 and Eq. 1.15).

$$\Gamma_m^L \approx \underbrace{2\pi \overline{|V_{1l}|^2} \rho_L(E_F)}_{\Gamma_1^L} |C_1^m|^2 \tag{1.14}$$

$$\Gamma_m^R \approx \underbrace{2\pi \overline{|V_{Nr}|^2} \rho_R(E_F)}_{\Gamma_N^R} |C_N^m|^2 \tag{1.15}$$

Eq. 1.14 and Eq. 1.15 demonstrate that the broadening of level m depends on the MO coefficients (C_1^m and C_N^m) at the electrode coupling sites (1 and N). To put it practically, if the orbital density at the linker groups is low then the transmission peak will be narrow, whereas if the orbital density is high then the peak will be broad.

In Eq. 1.14 and Eq. 1.15, the terms Γ_1^L and Γ_N^R describe the hopping rates onto site 1 and N of the molecule, respectively. These terms are often used in simplified tight-binding models, where the substitutions $\varepsilon_1 \to \varepsilon_1 - i\,\Gamma_1^L/2$ and $\varepsilon_N \to \varepsilon_N - i\,\Gamma_N^R/2$ are made to the on-site energies (ε) at

sites 1 and N. These substitutions indicate that the electron can leak into the left and right electrodes from sites 1 and N, respectively.

1.5.3 Transmission function for a single level

To calculate $T(E)$ from the golden rules, a relationship between $T(E)$ and the transition rates ($w_{i\to f}$) is needed. This can be achieved by expressing the current in terms of $w_{i\to f}$ and comparing it to the Landauer expression of current (Eq. 1.1). The modified golden rule (Eq. 1.10) can then be used to derive the transmission function for a single level.

The current flowing through a molecular junction can be described as a sum of ET rates. The current flowing from the left to the right, given by $I_{L\to R}$ (Eq. 1.16), is expressed as a sum over all ET rates $w_{l\to r}$ from occupied initial states $|l\rangle$ in the left electrode to final states $|r\rangle$ in the right electrode ($2e$ for spin degeneracy). A similar expression is given for $I_{R\to L}$ (Eq. 1.17).

$$I_{L\to R} = 2e \int dE f_L(E) \sum_l \delta(E - E_l) \sum_r w_{l\to r} \tag{1.16}$$

$$I_{R\to L} = 2e \int dE f_R(E) \sum_r \delta(E - E_r) \sum_l w_{r\to l} \tag{1.17}$$

The net current is the difference between the forward and reverse currents. If it is assumed that $w_{r\to l} = w_{l\to r}$ and that $w_{l\to r}$ is nonzero only if $E_l = E_r$ (i.e., elastic scattering), then the net current is given by Eq. 1.18.

$$I = I_{L\to R} - I_{R\to L} = 2e \int dE \big(f_L(E) - f_R(E)\big) \sum_l \delta(E - E_l) \sum_r w_{l\to r} \tag{1.18}$$

Comparing Eq. 1.18 to the Landauer formula (Eq. 1.1), $T(E)$ can be expressed as a sum of ET rates between pairs of states with energy E (Eq. 1.19).

$$T(E) = h \sum_l \delta(E - E_l) \sum_r w_{l\to r} \tag{1.19}$$

The transmission function for a single energy level E_m (Eq. 1.20) can be calculated by inserting the modified golden rule 1 (Eq. 1.10) and the definitions for $\Gamma_m^L(E)$ (Eq. 1.12) and $\Gamma_m^R(E)$ (Eq. 1.12) into Eq. 1.19.

$$\begin{aligned} T(E) &= h\sum_l \delta(E-E_l)\sum_r w_{l\to r} \\ &= 4\pi^2\sum_l \delta(E-E_l)\sum_r \left|\frac{V_{rm}V_{ml}}{E_l - E_m + i\,\Gamma_m(\mathrm{E}_l)/2}\right|^2 \delta(E_r - E_l) \\ &= \frac{\Gamma_m^L(E)\Gamma_m^R(E)}{|E - E_m + i\,\Gamma_m(E)/2|^2} \end{aligned} \tag{1.20}$$

Assuming Γ_m^L and Γ_m^R to be energy-independent (wide-band limit) and substituting $\Gamma_m = \Gamma_m^L + \Gamma_m^R$ into Eq. 1.20 yields the transmission function for a single level (Eq. 1.21). The transmission function has a Lorentzian line shape that is peaked at $E = E_m$ and has a width related to Γ_m. The level broadening is inversely related to the level lifetime ($\tau_m = \hbar/\Gamma_m$), which provides a convenient interpretation of the transmission function. Transmission of an electron with energy E occurs via energy non-conserving transitions to and from state $|m\rangle$. If the lifetime of the electron on state $|m\rangle$ is long, energy conservation is more stringent ($E \approx E_m$), and the transmission peak is narrow. In contrast, if the lifetime on state $|m\rangle$ is short, energy conservation can be relaxed and the transmission peak broadens. This phenomenon is a manifestation of the time-energy uncertainty principle.

$$T(E) = \frac{\Gamma_m^L\Gamma_m^R}{(E-E_m)^2 + (\Gamma_m^L + \Gamma_m^R)^2/4} \tag{1.21}$$

As a word of caution, although the Lorentzian line shape of $T(E)$ (Eq. 1.21) resembles that of the density of states for a single level, in general, the transmission function and the density of states are quite different. In particular, the transmission function is heavily influenced by quantum interference, which plays no role in the density of states. Thus, even though the single-level model

has a simple form (Eq. 1.21), a proper derivation requires some effort and nuanced discussion.

To summarize, the transfer of an electron across a molecular junction is facilitated by indirect transitions via the MOs of the molecule (per Fermi's golden rule 1). The indirect transition can be decomposed into two direct transitions with transition rates $\Gamma_m^L/\hbar$ and $\Gamma_m^R/\hbar$ (given by Fermi's golden rule 2). Since electron exchange occurs between the molecule and the electrodes, state $|m\rangle$ on the molecule can be viewed as a "virtual" state with energy $E_m - i(\Gamma_m^L + \Gamma_m^R)/2$, resulting in a broadening in the transmission peak. In the following sections, this notion of a virtual state with a complex energy will be further substantiated.

1.5.4 Generalized golden rule

In the prior section, the transmission function for a single level was calculated by making the substitution: $E_m \to E_m - i\,\Gamma/2$. In this section, the exact transition rate will be calculated, and the origin of the substitution will be clarified. The exact transition rate $w_{i\to f}$ from an initial state $|i\rangle$ in the left electrode to a final state $|f\rangle$ in the right electrode is given by a "generalized" golden rule expression (Eq. 1.22).[42] Eq. 1.22 is viewed as a generalized golden rule, because it takes the same form as Fermi's golden rules (Eq. 1.5 and Eq. 1.6), but with a different transition matrix element $\mathbb{T}_{fi}$.

$$w_{i\to f} = \frac{2\pi}{\hbar}|\mathbb{T}_{fi}|^2\delta\left(E_f - E_i\right) \tag{1.22}$$

The transition matrix element $\mathbb{T}_{fi}$ is given by Eq. 1.23. The first term of Eq. 1.23 (V_{fi}) is exactly the matrix element from golden rule 2 (Eq. 1.5), which describes direct transitions. The second summation, which describes indirect transitions, resembles the summation in golden rule 1 (Eq. 1.6) with one key difference. In golden rule 1, the summation ran over the *unperturbed* eigenstates of $\widehat{H}_0$, whereas for $\mathbb{T}_{fi}$ (Eq. 1.23), the summation runs over the *perturbed* eigenstates

$|n\rangle$, which are eigenstates of $\hat{H} = \hat{H}_0 + \hat{V}$ with energy E_n. Therefore, each term in the summation (Eq. 1.23) describes an indirect transition between the unperturbed states, $|i\rangle$ and $|f\rangle$, via the perturbed state $|n\rangle$. This simple tweak to golden rule 1—summing over the perturbed states instead of the unperturbed states—is enough to calculate the exact transition rate.

$$\mathbb{T}_{fi} = V_{fi} + \sum_n \frac{V_{fn}V_{ni}}{(E_i + i\eta) - E_n} \tag{1.23}$$

Also included in Eq. 1.23 is a small positive number, $\eta \to 0^+$, which protects the singularity at $E_i = E_n$. Technically this term should also have been included in golden rule 1 (Eq. 1.6). In deriving golden rule 1, the constant perturbation $\hat{V}$ is treated as a time-dependent perturbation $\hat{V}(t) = e^{\eta t}\hat{V}$ and the limit $\eta \to 0^+$ is taken at the end of the calculation to yield the constant perturbation result. This mathematical trick, known as the "slow-turn-on method", is needed to compute the integrals in the derivation.[43]

Even though Eq. 1.23 is exact, it may not seem very useful since the perturbed eigenstates $|n\rangle$ are unknown. If the perturbed states were known, there would be no need for perturbation theory. However, it is possible to calculate $\mathbb{T}_{fi}$ exclusively in terms of the unperturbed states (the collection of $|l\rangle, |m\rangle, |r\rangle$). To do this, it is useful define a matrix $\hat{G}(E)$ (Eq. 1.24), known as the Green's function. Using the Green's function (Eq. 1.24), $\mathbb{T}_{fi}$ is given by Eq. 1.25.

$$\hat{G}(E) = \lim_{\eta \to 0^+} \sum_n \frac{|n\rangle\langle n|}{(E + i\eta) - E_n} \tag{1.24}$$

$$\mathbb{T}_{fi} = V_{fi} + \left(\hat{V}\hat{G}(E_i)\hat{V}\right)_{fi} \tag{1.25}$$

As in the previous section, it is assumed that there is only a single level on the molecule, i.e., only one state $|m\rangle$. Hence, V_{ml}, V_{lm}, V_{mr},and V_{rm} (for all l and r) are the only matrix elements of $\hat{V}$ in the unperturbed basis that are nonzero (e.g., $V_{rl} = 0$). Therefore, $\mathbb{T}_{fi}$ (Eq. 1.25) simplifies

to equation Eq. 1.26 (as a reminder: i and f denote a specific l and r, respectively).

$$\mathbb{T}_{fi} = V_{fm} G_{mm}(E_i) V_{mi} \tag{1.26}$$

Applying Eq. 1.26 to the generalized golden rule (Eq. 1.22) and then following the transmission function derivation (Eq. 1.20), the transmission function can be expressed in the form of Eq. 1.27. Eq. 1.27 is the single-level version of the Caroli equation,[45-46] which is the general equation used to calculate $T(E)$.

$$T(E) = \Gamma_m^L(E) G_{mm}(E) \Gamma_m^R(E) G_{mm}^*(E) \tag{1.27}$$

1.5.5 Computing the Green's function

The Green's function matrix has simplified the calculation of $\mathbb{T}_{fi}$ (Eq. 1.26). The summation over the perturbed eigenstates $|n\rangle$ in Eq. 1.23 is replaced by a single matrix element $G_{mm}(E)$ (for the single-level model). Using some matrix manipulations, $G_{mm}(E)$ can be expressed in terms of the unperturbed eigenstates (the collection of $|l\rangle, |m\rangle, |r\rangle$). The Green's function satisfies Eq. 1.28, where $\hat{I}$ is the identity matrix, and the completeness relation has been invoked ($\hat{I} = \sum_n |n\rangle\langle n|$). Eq. 1.28 is typically the definition of $\hat{G}(E)$.

$$\left[(E + i\eta)\hat{I} - \hat{H}\right]\hat{G}(E) = \hat{I} \tag{1.28}$$

Eq. 1.28 can be expanded using $\hat{H} = \hat{H}_0 + \hat{V}$ (Eq. 1.4) to yield Eq. 1.29.

$$\left[(E + i\eta)\hat{I} - \hat{H}_0\right]\hat{G}(E) - \hat{V}\hat{G}(E) = \hat{I} \tag{1.29}$$

The matrix element $G_{mm}(E)$ can be calculated by applying the bra $\langle m|$ and ket $|m\rangle$ to Eq. 1.29 to give Eq. 1.30. In Eq. 1.30, the term $(\hat{V}\hat{G}(E))_{mm}$ has been expanded in the unperturbed basis (l, m, and r), but since $V_{mm} = 0$, the index k runs only over the electrode states (l and r). On the right-hand side of Eq. 1.30, normalization ($\langle m|m\rangle = 1$) is invoked.[47]

$$[(E + i\eta) - E_m] G_{mm}(E) - \sum_{k=l,r} V_{mk} G_{km}(E) = 1 \tag{1.30}$$

Eq. 1.30 expresses $G_{mm}(E)$ in terms of the elements $G_{km}(E)$. Each matrix element $G_{km}(E)$ can be calculated by applying bra $\langle k|$ and ket $|m\rangle$ to Eq. 1.29, yielding Eq. 1.31. The term $(\hat{V}\hat{G}(E))_{km}$ has been expanded in the unperturbed basis (l, m, and r), but V_{km} is the only nonzero element, so $(\hat{V}\hat{G}(E))_{km} = V_{km}G_{mm}(E)$. One right-hand side of Eq. 1.31: $\langle k|m\rangle = 0$.

$$[(E + i\eta) - E_k]G_{km}(E) - V_{km}G_{mm}(E) = 0 \tag{1.31}$$

The expression for $G_{km}(E)$ (Eq. 1.31) can be inserted into Eq. 1.30. This yields an expression for $G_{mm}(E)$ given by Eq. 1.32, where $\Sigma_m(E)$ (Eq. 1.33) is known as the self-energy.

$$G_{mm}(E) = \frac{1}{(E + i\eta) - E_m - \Sigma_m(E)} \tag{1.32}$$

The self-energy describes the effect that the electrodes have on the molecule due to electron exchange. The self-energy $\Sigma_m(E)$ (Eq. 1.33) bears striking resemblance to golden rule 1 (Eq. 1.6). Comparing the two equations suggests that $\Sigma_m(E)$ describes an indirect transition from state $|m\rangle$ back to itself via states in the electrodes (i.e., $w^{(2)}_{m\to m}$), hence the name self-energy.

$$\Sigma_m(E) = \sum_{k=l,r} \frac{V_{mk}V_{km}}{(E + i\eta) - E_k} \tag{1.33}$$

The self-energy $\Sigma_m(E)$ is a complex number (Eq. 1.34). The real and imaginary parts of $\Sigma_m(E)$ can be found using the Sokhotski–Plemelj relation, $\lim_{\eta\to 0^+} 1/(x + i\eta) = \mathcal{P}(1/x) - i\pi\delta(x)$, where $\mathcal{P}$ denotes the Cauchy principal value. The imaginary part of $\Sigma_m(E)$ is given by the level broadening $\Gamma_m(E)$ (Eq. 1.12–1.13). The real part of $\Sigma_m(E)$ is known as the level shift $\Lambda_m(E)$. $\Lambda_m(E)$ and $\Gamma_m(E)$ are related by a Hilbert transform (akin to a Fourier transform).

$$\Sigma_m(E) = \Lambda_m(E) - \frac{i}{2}\Gamma_m(E) \tag{1.34}$$

Inserting $G_{mm}(E)$ (Eq. 1.32) into Eq. 1.26 provides the transition matrix element ($\mathbb{T}_{fi}$) for a single level (Eq. 1.35). Eq. 1.35 demonstrates that golden rule 1 (Eq. 1.6) can be corrected

through the substitution $E_m \to E_m + \Sigma_m$. In the wide-band limit, Γ_m is energy-independent and the level shift (Λ_m) is zero (Γ_m and Λ_m are related). Thus, in the previous section, Λ_m was neglected so the substitution $E_m \to E_m - i\Gamma_m/2$ was made. This leads to the single-level model (Eq. 1.21).

$$\mathbb{T}_{fi} = \frac{V_{fm}V_{mi}}{(E_i + i\eta) - (E_m + \Sigma_m(E_i))} \tag{1.35}$$

A series expansion of $\mathbb{T}_{fi}$ provides an especially intriguing interpretation of the self-energy. $\mathbb{T}_{fi}$ can be expanded as a geometric series ($a(1-r)^{-1} = a + ar + ar^2 + \cdots$), yielding Eq. 1.36 (ignore the question of convergence). To simplify Eq. 1.36, the notation $\mathcal{E}_x = E_i + i\eta - E_x$ has been introduced (E_i is the energy of the initial/final states). The matrix elements of $\hat{V}$ in Eq. 1.36 suggest a satisfying interpretation of the series expansion. Each higher-order term represents an increasing number of virtual transitions made by an electron en route to the final state $|f\rangle$ from an initial state $|i\rangle$. The virtual transitions are highlighted above each term in Eq. 1.36. The additional virtual transitions in each successive term are provided by the self-energy. Far away from resonance, the first term in the series expansion (i.e., golden rule 1) may be sufficient, but close to resonance, the infinite series of terms is needed to properly calculate the transition rate.

$$\begin{aligned}
\mathbb{T}_{fi} &= \frac{V_{fm}V_{mi}}{\mathcal{E}_m}\left(1 - \frac{\Sigma(E_i)}{\mathcal{E}_m}\right)^{-1} \\
&= \frac{V_{fm}V_{mi}}{\mathcal{E}_m}\left(1 - \sum_{k=l,r}\frac{V_{mk}V_{km}}{\mathcal{E}_k\mathcal{E}_m}\right)^{-1} \\
&= \frac{\overbrace{V_{fm}V_{mi}}^{f\leftarrow m\leftarrow i}}{\mathcal{E}_m} + \sum_{k=l,r}\frac{\overbrace{V_{fm}V_{mk}V_{km}V_{mi}}^{f\leftarrow m\leftarrow l,r\leftarrow m\leftarrow i}}{\mathcal{E}_m\mathcal{E}_k\mathcal{E}_m} + \sum_{k=l,r}\sum_{k'=l',r'}\frac{\overbrace{V_{fm}V_{mk}V_{km}V_{mk'}V_{k'm}V_{mi}}^{f\leftarrow m\leftarrow l,r\leftarrow m\leftarrow l',r'\leftarrow m\leftarrow i}}{\mathcal{E}_m\mathcal{E}_k\mathcal{E}_m\mathcal{E}_{k'}\mathcal{E}_m} + \cdots
\end{aligned} \tag{1.36}$$

The derivation of the single-level model has hopefully demonstrated the utility of Fermi's golden rules in guiding one's intuition. At a basic level, the transmission function is described by

golden rule 1 (Eq. 1.6), which describes an electron undergoing an indirect transition from one electrode to the other via states on the molecule. The states on the single-molecule bridge have a modified energy $E_m + \Sigma_m$, where Σ_m is the self-energy. The self-energy (Σ_m) has the form of golden rule 1 (Eq. 1.6) and describes virtual transitions from the molecular state back to itself via states on the electrodes. The imaginary part of the self-energy (Eq. 1.34) describes leakage of electrons from the molecule (Eq. 1.11). The leakage can be viewed as a direct transition, described by golden rule 2 (Eq. 1.5). The leakage rate is equivalent to the hopping rate (described by Γ_m) that enters into the numerator of the single-level model. As such, the transmission function can be viewed as an elegant cooperation between Fermi's golden rule 1 and golden rule 2.

Chapter 2: Tight-binding analysis of helical states in carbyne

2.1 Preface

This chapter is adapted from the manuscript entitled *Tight-Binding Analysis of Helical States in Carbyne* by Suman Gunasekaran and Latha Venkataraman. This paper was inspired by a presentation given by Marc Garner at Columbia University.

2.2 Abstract

Carbyne is a linear allotrope of carbon that is composed of a chain of *sp*-hybridized carbon atoms. Through appropriate engineering of the chain termination, carbyne can harbor helical states where the π-electron delocalization twists along the axis of the chain. Herein, we present a comprehensive analysis of these helical states at the tight-binding level. We demonstrate that, in general, the molecular orbital coefficients of the helical states trace out an ellipse, in analogy to elliptically polarized light. Helical states can be realized in a model, inspired by the structure of cumulene, which considers a chain terminated by sp^2-hybridized atoms oriented at a nontrivial dihedral angle. We provide a complete analytic solution for this model. Additionally, we present a variation of the model that yields perfectly helical states that trace out a circle as opposed to an ellipse. Our results provide a deeper understanding of helical states and lay a foundation for more advanced levels of theory.

2.3 Introduction

Significant efforts have been made to engineer nanomaterials with designer electronic states.[48-51] Carbyne is a nanomaterial of interest that comprises a chain of *sp*-carbon atoms.[52-54] While polymeric carbyne has not yet been realized,[55-56] short carbyne chains are synthetically accessible as polyynes or cumulenes.[57-59] Each *sp*-carbon atom in the chain possesses two degenerate, mutually-perpendicular *p* orbitals that can be arbitrarily oriented.[60] This confers a

unique form of electron delocalization to carbyne that may be harnessed to realize novel electronic states. Unlike in conventional sp^2-carbon systems where the π-system has a fixed nodal plane, in carbyne the nodal plane can be oriented at any polar angle with respect to the axis of the chain. It has been shown that through appropriate engineering of the chain termination, carbyne can harbor helical states where the nodal plane of the π-molecular orbitals (MOs) rotates along the axis of the chain.[60-63] This *electrohelicity*[64] is associated with ring currents and antiresonances and may be useful for designing nanoscale electronics.[65-66] Herein, we provide a detailed analysis of the helical MOs in carbyne systems at the tight-binding level.

The existence of helical MOs in carbyne can be understood by considering an infinite chain where the chain termination is neglected. We will consider a carbyne chain oriented along the *z*-axis. The π-system of carbyne can be modeled with a basis set comprising a p_x and a p_y orbital for each *sp*-carbon atom in the chain. Note that the p_z orbitals are along the chain direction and therefore part of the σ-system and not the π-system. For the tight-binding model, we consider a coupling, $t < 0$, between adjacent p_x orbitals as well as adjacent p_y orbitals. The eigenstates, Ψ_k, and eigenenergies, E_k, associated with the wavenumber, k, are,[67-68]

$$\Psi_k(\mathrm{z}) = \begin{pmatrix} \psi_x(z) \\ \psi_y(z) \end{pmatrix} = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} e^{ikz} \tag{2.1}$$

$$E_k = 2t \cos k \tag{2.2}$$

where C_1 and C_2 are complex numbers, and z is the integer position of the atomic site along the chain. The components $\psi_x(z)$ and $\psi_y(z)$ represent the MO coefficients for the p_x and p_y orbitals on atom z, respectively. Since there is no coupling between ψ_x and ψ_y, i.e., $\langle \psi_y | H | \psi_x \rangle = 0$, the complex numbers C_1 and C_2 can be chosen arbitrarily. Mathematically, Eq. 2.1 is identical to the

Jones vector formulation for elliptically polarized light.[69-70] In analogy to light, if C_1 and C_2 differ by a complex phase, the eigenstate Ψ_k will be "elliptically polarized".

The heuristic explanation of helical states can be further developed by incorporating bond alternation into the tight-binding chain. Eq. 2.1 and Eq. 2.2 assume uniform coupling, t, which is only a valid approximation for even [n]cumulenes (an odd number of carbon atoms).[59, 71-72] For odd [n]cumulenes, polyynes, and polymeric carbyne, due to Peierls instability, the chain is more accurately modelled with alternating coupling, $t_1, t_2 < 0$.[73-75] With alternating coupling included, the infinite chain has a unit cell that comprises two atoms (i.e., sublattice A and B). We will consider the intracell coupling to be given by t_1 and the intercell coupling by t_2.[76] The eigenstates are described by a Bloch wave, as shown in Eq. 2.3, where *z* now indicates the integer position of the unit cell as opposed to the atomic site. The associated energy levels fall into a lower band (E_k^-) and an upper band (E_k^+) that are separated by a band gap (Eq. 2.4).[77-79]

$$\Psi_k^{A(B)}(z) = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} u_{A(B)} e^{ikz} \tag{2.3}$$

$$E_k^{\pm} = \pm\sqrt{t_1^2 + t_2^2 + 2t_1 t_2 \cos k} \tag{2.4}$$

Due to the discrete basis of p orbitals, the periodic component of the Bloch wave is described by discrete values u_A and u_B for sublattices A and B, respectively. The values of u_A and u_B are equal in magnitude ($|u_B| = |u_A|$) but differ in phase. For the lower band of states (i.e., E_k^-), $u_B = -e^{ik/2+\eta(k)} u_A$, and for the upper band of states (i.e., E_k^+), $u_B = e^{ik/2+\eta(k)} u_A$, where we have defined $\eta(k) = \arg\left[t_1 e^{-ik/2} + t_2 e^{ik/2}\right]$.[76, 80] Incorporating the phase difference between u_A and u_B into Eq. 2.3, we see that Ψ_k^A and Ψ_k^B are translationally offset in the *z*-direction, yielding a *double helix* structure: one helix for sublattice A and another for sublattice B. Since Ψ_k^A and Ψ_k^B

share the same Jones vector, $\begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$, these two helices will follow the same ellipse shape. Hence, the core features of the helical states are maintained even when bond alternation is included.

The above analysis demonstrates that carbyne can support helical states in principle. The challenge is to design finite chains with end groups that introduce a phase shift between $\psi_x(z)$ and $\psi_y(z)$. We present an analytic model based on cumulene[60] that provides an explicit example of how helical states can be achieved. For the sake of simplicity, we only consider uniform coupling, since the inclusion of alternating coupling would not affect the main results. In the final section, we present a generalized framework to analyze helical states in tight-binding chains that can be used to construct designer helical states.

2.4 Results and discussion

2.4.1 Linearly Polarized MOs

Before discussing the elliptically polarized MOs of the cumulene-based model, it is instructive to first consider finite sp^2-carbon chains where the MOs are "linearly polarized". For a finite chain of N coplanar p orbitals with uniform coupling t, the MOs are standing waves enumerated by quantum number n,[80-82]

$$\psi_n(z) = \sqrt{\frac{2}{N+1}} \sin k_n z \quad \text{for } 1 \leq z \leq N \tag{2.5}$$

The MO coefficients are a discrete sampling of a sine wave at integer values of z. As an example, the eigenstates for $N = 4$ (i.e., butadiene) are presented in Figure 2.1. Unlike for infinite chains where the wavenumber can freely vary, for finite chains the boundary conditions imposed by the chain termination yields quantized wavenumbers given as,

$$k_n = \frac{n\pi}{N+1} \quad \text{for } 1 \leq n \leq N \tag{2.6}$$

This quantization condition is akin to *particle-in-a-box*, where the electron is confined to a box of length $L = N + 1$.[83] The eigenenergies, E_n, associated with ψ_n exhibit the same functional form as for infinite chains (Eq. 2.2): $E_n = 2t \cos k_n$. In fact, Eq. 2.2 holds for all the systems discussed herein, although the specific wavenumbers, k_n, depend on the boundary conditions of the system.

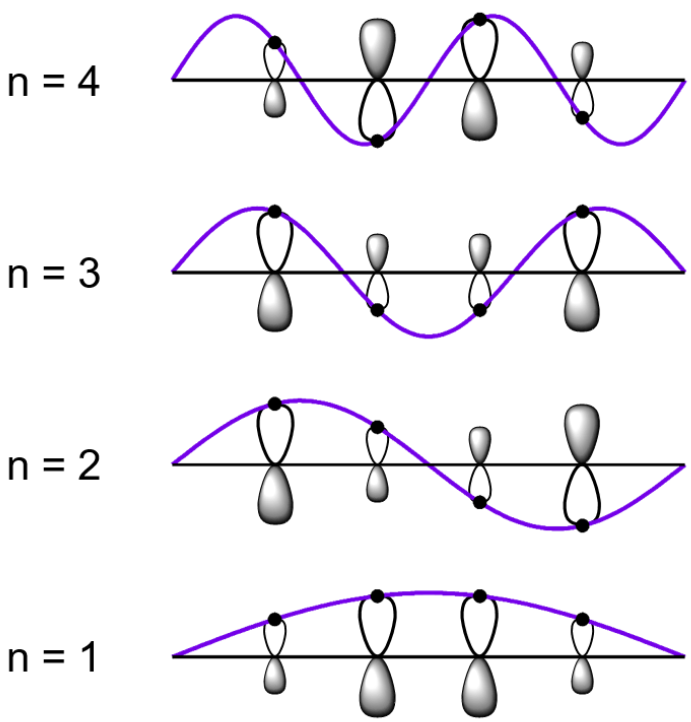

Figure 2.1. "Linearly polarized" MOs for a tight-binding chain of $N = 4$ p orbitals. The MO coefficients are indicated by the size of the p orbital and can be viewed as a discrete sampling of a sine wave (purple).

2.4.2 Elliptically Polarized MOs

To realize "elliptically polarized" states, a phase difference between ψ_x and ψ_y must be introduced using boundary conditions that asymmetrically couple ψ_x and ψ_y. A convenient method to produce this effect is to twist the ends of the chain.[62-63, 84-85] To explicitly construct helical MOs, we will consider a model that is motivated by the electronic structure of cumulene.[60]

The Hamiltonian, $H_N(\theta)$, describes a chain of $N-1$ *sp*-carbon atoms terminated with single *p* orbitals (i.e., sp^2-carbon atoms) oriented at a dihedral angle, θ (Figure 2.2a). Including the terminal sp^2-carbon atoms, the chain has a total of $N+1$ atoms and comprises a total of $2N$ *p* orbitals. This definition of N is consistent with that of n in [n]cumulene, which describes the number of cumulated double bonds.[59] The Hamiltonian is constructed using the resonance integral,

$$\left\langle p_\alpha^i \middle| p_\beta^{i+1} \right\rangle = t\cos(\alpha-\beta) \tag{2.7}$$

where α is the angle of a *p* orbital on atom $z=i$ and β is the angle of a *p* orbital on atom $z=i+1$. The Hamiltonian for $N=3$ is presented in Eq. 2.8 and Figure 2.2a.

$$H_3(\theta) = \begin{array}{c} \\ p_x^{(0)} \\ p_x^{(1)} \\ p_y^{(1)} \\ p_x^{(2)} \\ p_y^{(2)} \\ p_\theta^{(3)} \end{array} \begin{array}{c} \begin{array}{cccccc} p_x^{(0)} & p_x^{(1)} & p_y^{(1)} & p_x^{(2)} & p_y^{(2)} & p_\theta^{(3)} \end{array} \\ \begin{pmatrix} 0 & \boldsymbol{t} & 0 & 0 & 0 & 0 \\ \boldsymbol{t} & 0 & 0 & \boldsymbol{t} & 0 & 0 \\ 0 & 0 & 0 & 0 & \boldsymbol{t} & 0 \\ 0 & \boldsymbol{t} & 0 & 0 & 0 & \boldsymbol{t\cos\theta} \\ 0 & 0 & \boldsymbol{t} & 0 & 0 & \boldsymbol{t\sin\theta} \\ 0 & 0 & 0 & \boldsymbol{t\cos\theta} & \boldsymbol{t\sin\theta} & 0 \end{pmatrix} \end{array} \tag{2.8}$$

Since the chain comprises $N+1$ atoms, for $H_3(\theta)$ there are 4 atoms labeled as $z=0,1,2,3$. The first atom in the chain ($z=0$) is sp^2-hybridized and provides only one *p* orbital to the π-system. Without loss of generality, this *p* orbital is oriented along the *x*-axis. The last atom in the chain ($z=N$) is also sp^2-hybridized, with the lone *p* orbital oriented at an angle θ from the *x*-axis. This dihedral angle between the terminal *p* orbitals will be referred to as the twist angle. The atoms in the core of the chain are *sp*-hybridized and are described by two perpendicular *p* orbitals that are oriented along the *x*- and *y*-axes, respectively (Figure 2.2a).

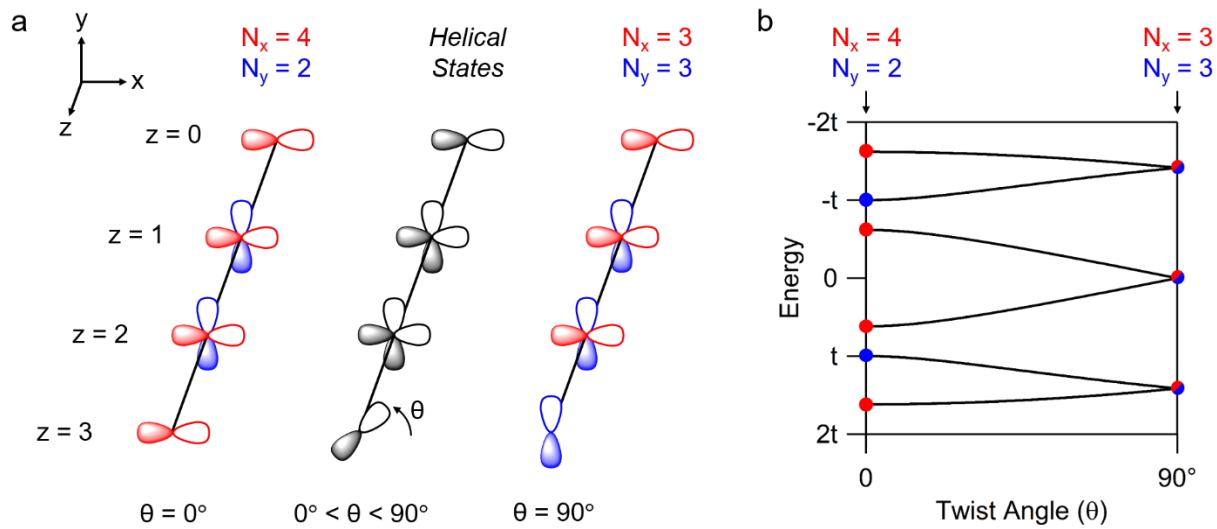

Figure 2.2. a) Visual representation of Hamiltonian $H_N(\theta)$ for N = 3. The coupling between adjacent p orbitals is given by Eq. 2.7. Note that the p_z orbitals involved in the σ-system are not shown. For θ = 0° and θ = 90°, H_N can be partitioned into two subsystems indicated in red and blue. b) Energy spectrum for $H_N(\theta)$ for N = 3. At θ = 0° and θ = 90°, the energy levels for the two subsystems are indicated in red and blue, corresponding to (a).

The energy levels of H_N are given by the roots of the characteristic polynomial, $\det(H_N - EI)$, where I denotes the identity matrix. The characteristic polynomial takes a simple form if the energy, E, is expressed as $E_k = 2t\cos k$ (Eq. 2.2):[86-87]

$$\det(H_N - E_k I) = t^{2N}\left(\frac{\sin^2(N+1)k}{\sin^2 k} - \cos^2\theta\right) \tag{2.9}$$

Setting Eq. 2.9 to zero yields a trancendental equation describing the allowed k_n for a given twist angle θ,

$$\sin^2(N+1)k_n = \cos^2\theta\sin^2 k_n \quad \text{for } 1 \le n \le 2N \tag{2.10}$$

There are two limiting cases of Eq. 2.10 that are worth noting: θ = 0° and θ = 90°. In both situations, H_N can be partitioned into two uncoupled π-systems indicated by red and blue in Figure 2.2a. The eigenstates for these uncoupled π-systems are "linearly polarized" (e.g., Figure 2.1). Accordingly, Eq. 2.10 reduces to *particle-in-a-box* quantization (Eq. 2.6). For $\theta = 0°$, Eq. 2.10 can be re-expressed as $\sin(N+2)k_n \sin Nk_n = 0$, which yields the condition of Eq. 2.6 for chains

of length $N+1$ and $N-1$. Similarly, for $\theta = 90°$, Eq. 2.10 reduces to $\sin^2(N+1)k_n = 0$, which describes two chains of length N (Figure 2.2a). The limiting cases manifest in the energy spectrum for $H_3(\theta)$ presented in Figure 2.2b. At θ = 0° and θ = 90°, the energy levels correspond to the energy levels of Hückel models for standard *sp*^{2}-carbon molecules. At $\theta = 0°$, the energy levels correspond to butadiene ($N_x = 4$, red) and ethylene ($N_y = 2$, blue), while at $\theta = 90°$, the energy levels are doubly degenerate and correspond to the allyl radical ($N_x = N_y = 3$).

Between the two limiting cases mentioned above, i.e., for nontrivial twist angles: $0° < \theta < 90°$, the energy levels correspond to helical states where the p_x and p_y π-systems are inextricably linked. In Figure 2.3, the helical states of $H_3(\theta = 60°)$ are presented. Unlike Figure 2.1, where the MOs were depicted from the side (*z*-axis pointing to the right), in Figure 2.3, the MOs are depicted from the front, with the positive *z*-axis pointing out of the page. Following from the definition of H_N (Figure 2.2a), the MO coefficients at $z = 0$ and $z = N$ are depicted on the *x*- and *θ*-axes, respectively. The MO coefficients for the core of the chain, $1 \leq z \leq N-1$, are plotted in vector form, $\Psi_n(\mathrm{z}) = \begin{pmatrix} \psi_x(z) \\ \psi_y(z) \end{pmatrix}$, and can be arbitrarily oriented. Figure 2.3 demonstrates that, for $0° < \theta < 90°$, the MOs are no longer confined to single plane as in Figure 2.1. Instead, the MOs are "elliptically polarized" and can be parametrized in the form,

$$\Psi_n(\mathrm{z}) = \begin{pmatrix} \cos\theta/2 & -\sin\theta/2 \\ \sin\theta/2 & \cos\theta/2 \end{pmatrix} \begin{pmatrix} a_n \cos(k_n z + \delta_n) \\ b_n \sin(k_n z + \delta_n) \end{pmatrix} \tag{2.11}$$

where a_n and b_n are the semi-major/minor axes of the ellipse. The left matrix in Eq. 2.11 is a 2x2 rotation matrix that rotates the ellipse to an orientation angle of $\phi = \theta/2$. This orientation angle follows from the symmetry of the chain. An additional phase factor, δ_n, is needed so that $\psi_y(0) = 0$, since the p_x orbital at $z = 0$ cannot have a ψ_y component. This condition is satisfied by the formula: $\tan\delta_n = -a_n/b_n \tan(\theta/2)$.

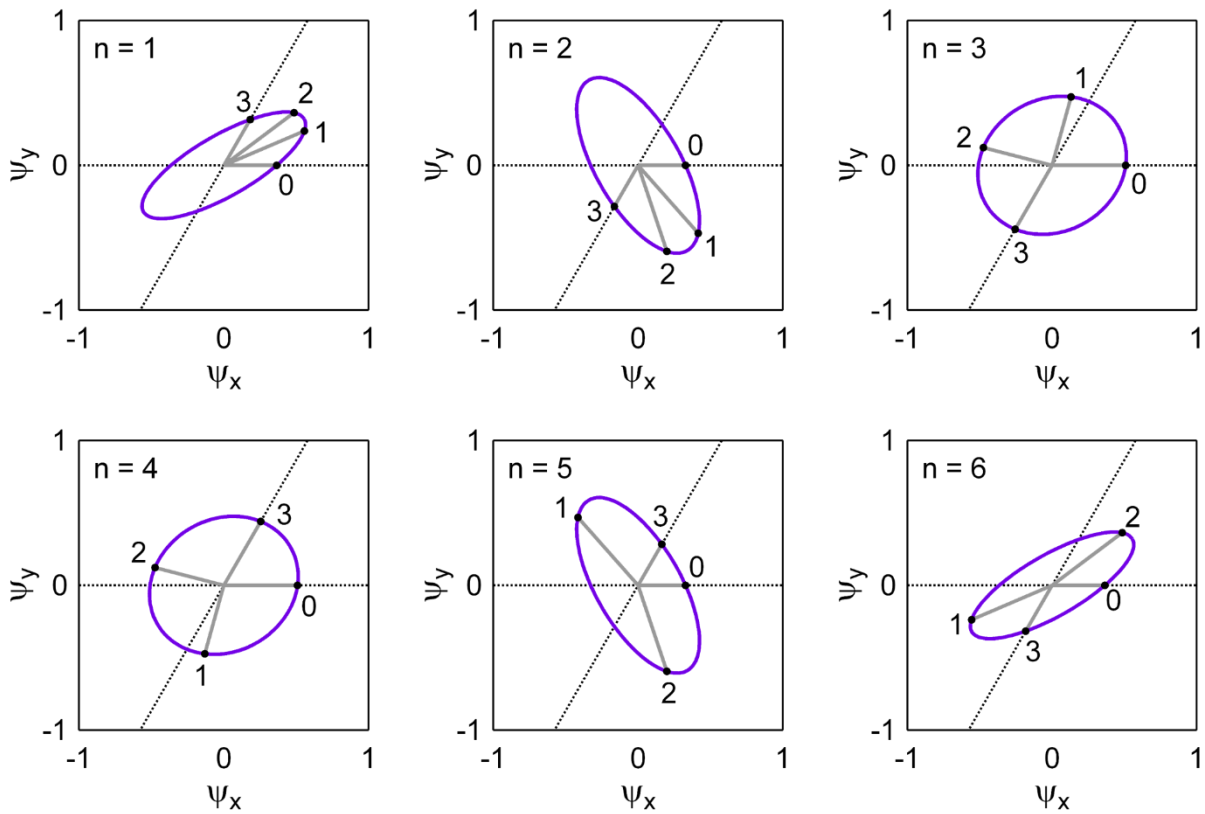

Figure 2.3. The helical MOs of $H_N(\theta)$ for N = 3 and θ = 60°. The MOs are plotted head-on, with the positive z-axis pointing out of the page. The MO coefficients follow an elliptical path (purple) around the axis of the chain. The handedness of the helix alternates between right-handed ($n = 1,3,5$) and left-handed ($n = 2,4,6$). The x- and θ-axes are indicated by the dashed lines.

To characterize the helical states, it is convenient to introduce an ellipticity angle, $-\pi/2 < \varphi < \pi/2$, where the semi-major/minor axes are given by $a = r\cos\varphi$ and $b = r\sin\varphi$ (r is determined by the normalization of the MO). When $|\varphi| < \pi/4$, a is the major axis, whereas when $|\varphi| > \pi/4$, b is the major axis (Figure 2.4). The sign of φ dictates the handedness of the helical state, with positive and negative φ yielding right-handed and left-handed helices, respectively. We derive the following formula for the ellipticity angle φ_n of the eigenstates Ψ_n,

$$\varphi^{\pm}(k) = \pm\frac{1}{2}\cos^{-1}\left(\frac{\pm\cos k}{\sqrt{1-\cos^2\theta\sin^2 k}}\right) \tag{2.12}$$

where $\varphi_n = \varphi^+(k_n)$ when *n* is odd, and $\varphi_n = \varphi^-(k_n)$ when *n* is even. The functions $\varphi^+(k)$ and $\varphi^-(k)$ are plotted in Figure 2.4b. Remarkably, these two functions do not depend on N and only depend on the properties of the chain termination, i.e., the twist angle (θ). For a specific N, the allowed wavenumbers k_n are given by Eq. 2.10 and can be determined graphically (Figure 2.4a). The ellipticity angles φ_n can then be calculated from the k_n as indicated in Figure 2.4b. In this way, the length of the chain only determines the specific φ_n but not the overall trends of $\varphi^{\pm}(k)$.

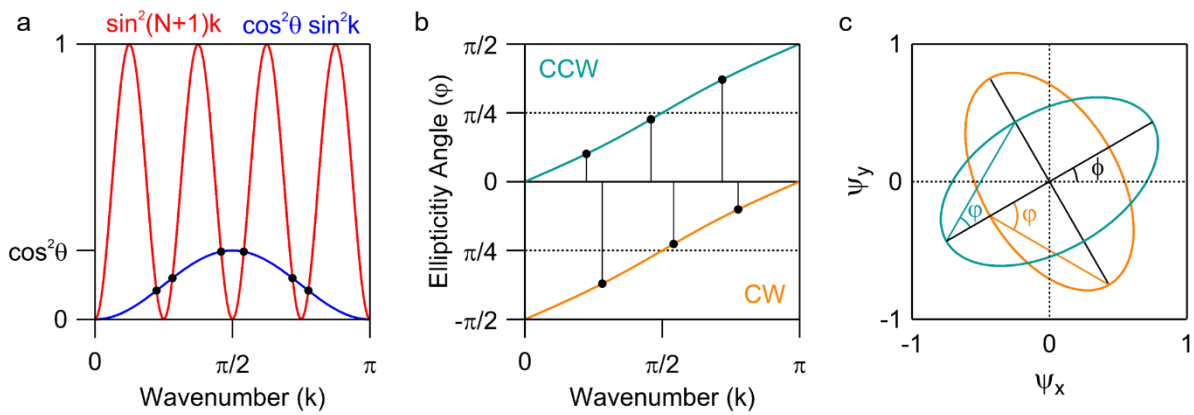

Figure 2.4.Mathematical description of helical states of $H_N(\theta)$ for $N = 3$ and $\theta = 60°$. a) Graphical representation of Eq. 2.10. The allowed wavenumbers k_n are given by the points of intersection. b) The ellipticity angle as a function of wavenumber Eq. 2.12, given by $\varphi^+(k)$ (teal) and $\varphi^-(k)$ (orange). The specific φ_n for $N = 3$ are indicated by the black markers and correspond to the points of intersection in (a). The vertical sticks are included to guide the eye. c) The ellipses described by $\varphi^+(k)$ (teal) and $\varphi^-(k)$ (orange) for $k = 0.35\pi$. The ellipses are oriented at an angle $\phi = \theta/2$.

Since $\varphi^+(k)$ is positive and $\varphi^-(k)$ is negative, these ellipticity angles describe right-handed and left-handed helices, respectively. Therefore, the handedness of the helical states alternates with increasing *n* (e.g., Figure 2.3). For a given k, $\varphi^+(k)$ and $\varphi^-(k)$ differ by $\pi/2$ and produce two perpendicularly oriented ellipses (Figure 2.4c). Lastly, we note that $\varphi^-(\pi - k) = -\varphi^+(k)$, which is a manifestation of the Coulson-Rushbrooke pairing theorem.[88] This pairing phenomenon can be seen in Figure 2.3 for the MO pairs: $n = 1,6$; $n = 2,5$; and $n = 3,4$. For each

MO pair (e.g., $n = 1,6$), both MOs follow the same ellipse shape but with opposite handedness (e.g., $n = 1$ is right-handed and $n = 6$ is left-handed).

2.4.3 Circularly Polarized MOs.

A slight modification to H_N (Eq. 2.8) simplifies the mathematics considerably and yields "circularly polarized" states. In Eq. 2.13 we consider a modified Hamiltonian, H'_N, where the coupling to the terminal lone *p* orbitals is increased to $\sqrt{2}t$ (Figure 2.5a).

$$H'_3(\theta) = \begin{array}{c} \\ p_x^{(0)} \\ p_x^{(1)} \\ p_y^{(1)} \\ p_x^{(2)} \\ p_y^{(2)} \\ p_\theta^{(3)} \end{array} \begin{array}{c} \begin{array}{cccccc} p_x^{(0)} & p_x^{(1)} & p_y^{(1)} & p_x^{(2)} & p_y^{(2)} & p_\theta^{(3)} \end{array} \\ \begin{pmatrix} 0 & \boldsymbol{\sqrt{2}t} & 0 & 0 & 0 & 0 \\ \boldsymbol{\sqrt{2}t} & 0 & 0 & \boldsymbol{t} & 0 & 0 \\ 0 & 0 & 0 & 0 & \boldsymbol{t} & 0 \\ 0 & \boldsymbol{t} & 0 & 0 & 0 & \boldsymbol{\sqrt{2}t\cos\theta} \\ 0 & 0 & \boldsymbol{t} & 0 & 0 & \boldsymbol{\sqrt{2}t\sin\theta} \\ 0 & 0 & 0 & \boldsymbol{\sqrt{2}t\cos\theta} & \boldsymbol{\sqrt{2}t\sin\theta} & 0 \end{pmatrix} \end{array} \tag{2.13}$$

The characteristic polynomial for H'_N is presented in Eq. 2.14 where the substitution $E = E_k$ (Eq. 2.2) has been made.

$$\det(H'_N - E_k I) = 2t^{2N}(\cos(2Nk) - \cos(2\theta)) \tag{2.14}$$

The roots of the characteristic polynomial give the allowed wavenumbers, k_n, listed as follows for even and odd n,

$$Nk_n = \begin{cases} \theta, \theta + \pi, \theta + 2\pi, \ldots & \text{for } n = 1,3,5,\ldots \\ \pi - \theta, 2\pi - \theta, 3\pi - \theta, \ldots & \text{for } n = 2,4,6,\ldots \end{cases} \tag{2.15}$$

Remarkably, the eigenstates for H'_N are "circularly polarized" and are given as,

$$\Psi_n(z) = \frac{1}{\sqrt{N}} \begin{pmatrix} \cos(k_n z) \\ \pm\sin(k_n z) \end{pmatrix} \qquad 1 \le z \le N - 1 \tag{2.16}$$

where (+) and (−) describe odd and even n, respectively. As before, the handedness of the helix alternates with increasing n. In Figure 2.5b,c, the frontier MOs for $H'(\theta = 60°)$ for $N = 6$ are presented. The MO coefficients of the core atoms ($1 \leq z \leq 5$) uniformly twist around the axis of the chain, following a circular path. However, due to the modified coupling, the terminal p orbitals do not obey Eq. 2.16, and the MO coefficients for $z = 0$ and $z = N$ are a factor of $1/\sqrt{2}$ lower in magnitude.

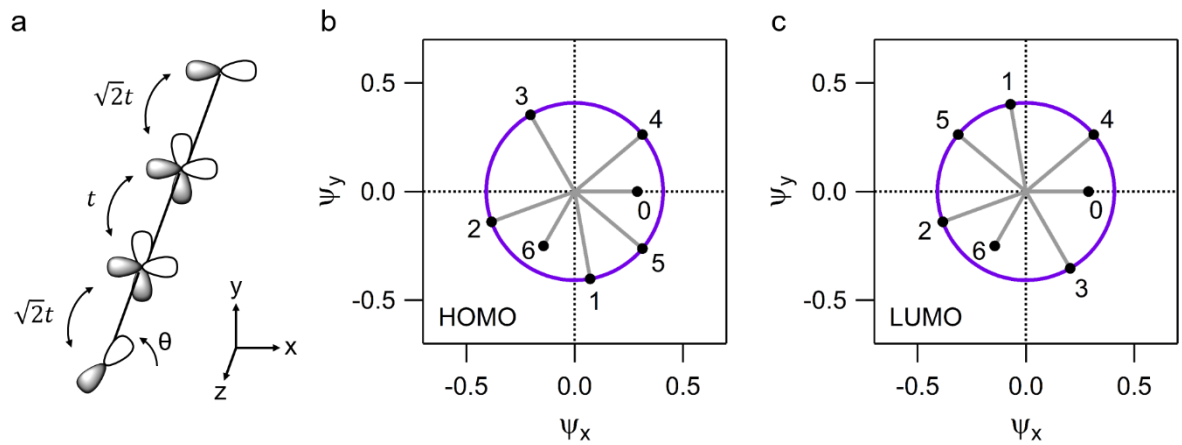

Figure 2.5. a) Visual representation of Hamiltonian $H'_N(\theta)$ for $N = 3$. b,c) Circularly polarized states of $H'_N(\theta = 60°)$ for a seven-carbon chain ($N = 6$) for (b) $n = 6$ (HOMO) and (c) $n = 7$ (LUMO). Note that we present a longer chain length in (b,c) than in (a) to better demonstrate the circular polarization.

2.4.4 Generalized Analysis

In the above cumulene-based model, helical states arise from a twist angel between terminal p orbitals. The simple chain termination, i.e., lone p orbitals at $z = 0$ and $z = N$, allowed us to derive a complete analytic description of the helical states. However, for more complicated chain terminations, which may better model realistic molecules,[65] analytic expressions cannot be obtained, and a more general framework is needed to analyze the helical states. In the following discussion, we generalize the results from the previous sections. We analyze the eigenstates of a finite carbyne chain that is terminated by arbitrary end groups. As we will show, the eigenstates

for any tight-binding *sp*-carbon chain will be elliptically polarized, with the familiar linearly polarized states only being a limiting case.

For the generalized analysis, it is convenient to invoke a direct product notation where the p_x orbital on atom *i* is described by $\begin{pmatrix}1\\0\end{pmatrix} \otimes |i\rangle$ and the p_y orbital on atom *i* is described by $\begin{pmatrix}0\\1\end{pmatrix} \otimes |i\rangle$. We will consider an *sp*-carbon chain of *N* atoms that is capped by some arbitrary end groups. Using the Löwdin partitioning technique,[89] the full Hamiltonian can be reduced to an effective Hamiltonian H of dimension $2N \times 2N$ (Eq. 2.17).

$$H = \Sigma_L(E) \otimes |1\rangle\langle 1| + \Sigma_R(E) \otimes |N\rangle\langle N| + \sum_{z=1}^{N-1} t\, I \otimes |z\rangle\langle z+1| + H.c. \tag{2.17}$$

The left (L) and right (R) end groups are incorporated into H by means of 2×2 matrices $\Sigma_L(E) = V_L(E - H_L)^{-1}V_L^{\dagger}$ and $\Sigma_R(E) = V_R\,(E - H_R)^{-1}V_R^{\dagger}$, where $H_{L(R)}$ is the Hamiltonian of the end group and $V_{L(R)}$ is the coupling between the end group and the terminal atoms of the chain.[90-91] In order for H to yield helical states, it is necessary that Σ_L and Σ_R do not commute. Otherwise Σ_L and Σ_R could be simultaneously diagonalized, and H could be separated into two uncoupled subsystems.

The eigenstates of H can be computed using an ansatz that describes an elliptically polarized wavefunction,

$$\Psi(z) = MR(kz)\hat{v}_0 \tag{2.18}$$

where $\hat{v}_0$ is some unit vector, R is the 2×2 rotation matrix, and M is a real, symmetric matrix characterizing the elliptical polarization. The matrix M posseses a pair of orthonormal eigenvectors $\hat{v}_1$ and $\hat{v}_2$ and associated eigenvalues a and b. The unit vector, $R(kz)\hat{v}_0$, rotates along the unit circle for increasing z and is mapped by M onto an ellipse whose major/minor axes are given by $a\hat{v}_1$ and $b\hat{v}_2$.[92-93] The handedness of the helical state is given by the sign of $\det[M] = ab$. For

right-handed helices, $\det[M]$ is positive, whereas for left-handed helices, $\det[M]$ is negative. We will consider the normalization condition: $a^2 + b^2 = 1$, which is equivalent to the condition $\mathrm{tr}[M^{\mathrm{T}}M] = 1$. These constraints leave M with two free parameters that can be the orientation angle, ϕ, and the ellipticity angle, φ,

$$M = R(\phi)\begin{pmatrix}\cos\varphi & 0 \\ 0 & \sin\varphi\end{pmatrix}R(-\phi) \tag{2.19}$$

For the ansatz (Eq. 2.18) to describe the eigenstates of H, the ansatz must satisfy,

$$E\Psi(z) = \langle z|H|\Psi\rangle \tag{2.20}$$

for all z. Within the bulk of the chain ($2 \leq z \leq N-1$), Eq. 2.20 yields,

$$\begin{aligned} E\Psi(z) &= t\Psi(z-1) + t\Psi(z+1) \\ &= tM[R(-k) + R(k)]R(kz)\hat{v}_0 \\ &= 2t\cos k\, MR(kz)\hat{v}_0 \end{aligned} \tag{2.21}$$

where we have made use of the relation: $R(-k) + R(k) = 2\cos k\, I$. Therefore, the ansatz correctly describes eigenstates of H with eigenenergies $E_k = 2t\cos k$ (Eq. 2.2). Eq. 2.21 demonstrates that, in general, energy levels for which $|E_n| \leq 2t$ will be associated with helical eigenstates. It should be noted that for certain $\Sigma_{L(R)}$, there can exist energy levels $|E_n| > 2t$ which cannot be satisfied by Eq. 2.21. In such instances, the associated eigenstates will not be helical.

Eq. 2.20 must also be satisfied at the ends of the chain. Application of Eq. 2.20 to $z = 1$ and $z = N$ yields Eq. 2.22 and Eq. 2.23, respectively.

$$\begin{aligned} E_k\Psi(1) &= \Sigma_L(E_k)\Psi(1) + t\Psi(2) \\ &= [\Sigma_L(E_k)M + tMR(k)]R(k)\hat{v}_0 \end{aligned} \tag{2.22}$$

$$\begin{aligned} E_k\Psi(N) &= t\Psi(N-1) + \Sigma_{\mathrm{R}}(E_k)\Psi(N) \\ &= [tMR(-k) + \Sigma_{\mathrm{R}}(E_k)M]R(Nk)\hat{v}_0 \end{aligned} \tag{2.23}$$

Substituting the relation $E_k M = tM\big(R(-k) + R(k)\big)$ into the left-hand side of Eq. 2.22 and Eq. 2.23 leads to the simplified Eq. 2.24 and Eq. 2.25, respectively.

$$[tMR(-k) - \Sigma_L(E_k)M]R(k)\hat{v}_0 = 0 \tag{2.24}$$

$$[tMR(k) - \Sigma_{\mathrm{R}}(E_k)M]R(Nk)\hat{v}_0 = 0 \tag{2.25}$$

It follows that the operator in the square brackets for both Eq. 2.24 and Eq. 2.25 has a nonzero kernel and consequently a zero determinant,

$$\det[tMR(-k) - \Sigma_L(E_k)M] = 0 \tag{2.26}$$

$$\det[tMR(k) - \Sigma_{\mathrm{R}}(E_k)M] = 0 \tag{2.27}$$

Eq. 2.26 and Eq. 2.27 allow us to determine the relationship between M and $\Sigma_{L(R)}$ without concerning ourselves with the specifics of Ψ_n. For a given value of k, these equations represent a system of quadratic equations that can be solved for $M(k)$. We find, in general, four solutions to these equations, two of which are unique. The two solutions can be expressed in the form,

$$M^+(k) = R(\phi)\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}R(-\phi) \quad \text{and} \quad M^-(k) = R(\phi)\begin{pmatrix} -b & 0 \\ 0 & a \end{pmatrix}R(-\phi) \tag{2.28}$$

The matrices $M^{\pm}(k)$ represent helical states of opposite handedness, for which the associated ellipses are perpendicularly oriented (i.e., Figure 2.4c). This is a nice generalization of $\varphi^{\pm}(k)$ (Eq. 2.12). In accordance with $\varphi^{\pm}(k)$, $M^{\pm}(k)$ do not depend on N and solely depend on the properties of the chain termination ($\Sigma_{L(R)}$). One can use Eq. 2.26 and Eq. 2.27 to engineer designer helical states through appropriate selection of $\Sigma_{L(R)}$.

In conclusion, we have demonstrated that the eigenstates of *sp*-carbon chains are "elliptically polarized," drawing analogy to elliptically polarized light. Helical states emerge when the ends of the chain are asymmetrically coupled, which can be made mathematically precise with the condition that $[\Sigma_L, \Sigma_R] \neq 0$. The relationship between the helical states and the wavenumber (i.e., $M^{\pm}(k)$) depends solely on the chain termination and not on the chain length. The chain length

only determines the specific allowed wavenumbers k_n. These ideas are made more concrete within a cumulene-based model, for which we provide the complete analytical solution. We hope the underlying patterns identified herein, at the tight-binding level, will inform future theoretical and experimental efforts into the electronic structure of carbyne.

Chapter 3: Length-dependent conductance of polymethines

3.1 Preface

This chapter is adapted from the manuscript entitled *Near length-independent conductance in polymethine wires* by Suman Gunasekaran, Daniel Hernangómez-Pérez, Iryna Davydenko, Seth Marder, Ferdinand Evers, and Latha Venkataraman.[94]

3.2 Abstract

Polymethine dyes are linear π-conjugated compounds with an odd number of carbons that display a much greater delocalization in comparison to polyenes that have an even number of carbon atoms in their main chain. Herein, we perform scanning tunneling microscope-based break-junction measurements on a series of three cyanine dyes of increasing length and demonstrate, at the single-molecule level, that these short chain polymethine systems exhibit a substantially smaller decay in conductance with length (attenuation factor $\beta = 0.04$ Å^{-1}) compared to traditional polyenes ($\beta \approx 0.2$ Å^{-1}). Further, we show that by changing solvent we are able to shift the β value, demonstrating a remarkable negative β value, with conductance increasing with molecular length. First principle calculations provide support for the experimentally observed near-uniform length dependent conductance and further suggest that the variations in β with solvent are due to solvent-induced changes in the alignment of the frontier molecular orbitals relative to the Fermi energy of the leads. Moreover, analysis of a simplified Hückel model suggests that the smaller decay in conductance correlates with the smaller degree of bond order alternation present in polymethine compounds compared to polyenes. These findings may lead to the design of true molecular wires that enable efficient electron transport without a length-dependent decay at the nanoscale.

3.3 Introduction

Significant efforts have been made to develop molecular wires capable of efficiently transporting charge at the nanometer scale.[95-97] Indeed, molecular wires are of interest for molecular electronics[98-99] as well as for donor-bridge-acceptor (D-B-A) systems[100-101] designed for photovoltaics or artificial photosynthesis. Most frequently, conjugated organic systems are explored because they comprise extended π-systems with significant electron delocalization. However, within the coherent and off-resonant transport regime, these molecular wires tend to suffer from an exponential attenuation of conductance (G) with wire length (L), i.e., $G \sim e^{-\beta L}$. This exponential decay trend is well documented in both molecular conduction measurements[21-22, 95, 98-99, 102-105] and optical measurements of D-B-A systems[100-101, 106] and has been derived from a variety of models.[107-111]

While the conductance of a molecular wire is strongly dependent on the nature of the contact between the molecular wire and the electrodes, the β value typically provides more general insight into the electronic properties of the molecular backbone. Further, the β value allows for reasonable comparisons to be made between molecular wires comprising different oligomers,[112] where systems with small β values are desirable for long-range electron transport applications. Polyenes, relevant herein, have reported β values of 0.22 Å^{-1},[102-103] 0.17 Å^{-1},[113] and polyene-like systems, such as thiophenes (β = 0.16 Å^{-1},[114] 0.4 Å^{-1} [115]) and thiophene dioxide oligomers (β = 0.20 Å^{-1} [116]) have β values similar to polyenes. While there is certainly variability in the reported values, polyene-like systems generally display $\beta \approx 0.2$ Å^{-1}.

Herein, in an effort to achieve a significantly smaller β, we sought to explore novel forms of π-conjugation and identified polymethine compounds to be of interest due to their unique electronic structure. Polymethines are linear conjugated chains comprising an odd number of sp^2

hybridized carbon atoms, in contrast to polyenes, which contain an even number.[117-118] Notably, unlike polyenes, symmetric polymethines display near-uniform π-bond orders along the chain.[119] This is in part an even-odd effect where, for short chains, the energetic gain from dimerization for an odd number of p-orbitals is significantly lower than for an even number. Accordingly, polymethine systems experimentally exhibit a smaller degree of bond length alternation.[117, 120-121] One can reason these effects from resonance structures where the single and double bonds are interchanged (Figure 3.1a). Due to this increased π-electron delocalization, in 1990 Reimers and Hush predicted that polymethines might exhibit a lower conductance attenuation with length than observed for polyenes.[122]

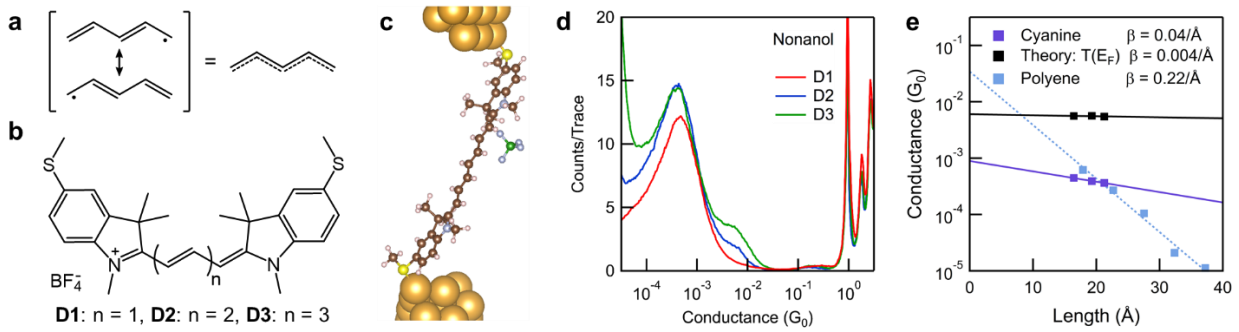

Figure 3.1. Single molecule conductance measurements of cyanine series. (a) Resonance structures of a polymethine radical, which suggest that polymethine systems have near uniform π bond orders. (b) Molecular structures of the cyanine dyes in the series studied. (c) Schematic representation of STM-BJ technique for measuring the conductance of single molecules. Pictured is the junction structure used to compute the transmission spectrum for **D3** in Figure 3.3a. (d) Logarithmically-binned histograms of 10,000 traces of measurements at 90 mV of 0.1 mM **D1-D3** in 1-nonanol. Histogram has 100 bins/decade (e) Length dependent conductance of cyanine compared to polyene.[103] The lengths of the cyanine wires were determined from density functional theory (DFT)-optimized structures as detailed in the text. All straight lines are least-square fits to the data. Cyanine exhibits a substantially smaller decay in conductance with wire length compared to polyene, in agreement with DFT-based transmission calculations (black squares).

3.4 Results and discussion

Remarkably, while numerous even carbon systems have been examined,[113-116] no single-molecule conductance measurements have been performed on polymethine derivatives. Herein, we probe for the first time the length-dependent electronic properties of a series of three cyanine dyes (labeled **D1**, **D2**, and **D3**, see Figure 3.1b), a common class of polymethine compounds.[123] Standard procedures and techniques were employed for the synthesis and characterization of the compounds.[94] The single-molecule conductance of **D1-D3** was measured using the scanning tunneling microscope break-junction technique (STM-BJ)[21] method as described previously.[22] Briefly, a solution containing the molecule of interest was deposited onto a gold-coated mica or steel substrate. A cut gold tip was submerged in the solution and positioned in proximity to the substrate. For each measurement, the Au tip was brought into contact with the substrate until a conductance greater than 5 G_0 was achieved ($G_0 = 2\ e^2/h$). As the tip was retracted, an atomic point contact was formed and subsequently broken, yielding a small gap, which was bridged by the molecule via gold-binding thioanisole linker groups (Figure 3.1c). The conductance (current/voltage) was recorded during the retraction, and the molecular conductance was identified by a plateau-like feature below 1 G_0 in the conductance versus displacement trace.

We performed STM-BJ measurements on 0.1 mM solutions of **D1-D3** in 1-nonanol. Ten thousand traces at a 90 mV bias were collected for each molecule and compiled into logarithmically binned histograms shown in Figure 3.1d. We observed a distribution of molecular conductance values with the peak positions representing the most probable molecular conductance. The peak position was determined using a Gaussian fit to the data, and these values are plotted against molecule length in Figure 3.1e on a semi-logarithm scale, with the lengths measured as the through-space distance between the two sulfur atoms obtained from density functional theory

(DFT) optimized structures. We fit these data to an exponential in order to extract a decay constant $\beta = 0.04 \pm 0.02$ Å^{-1}. We observe that this decay is considerably lower than the decay constant for polyenes ($\beta \approx 0.2$ Å^{-1}) as shown in Figure 3.1e. Alternatively, one can calculate β per C=C unit; we find $\beta = 0.10 \pm 0.04$ unit^{-1} for **D1-D3** compared to $\beta = 0.53$ unit^{-1} for polyene.[103]

Next, we perform measurements in 1,2,4-trichlorobenzene (TCB) and propylene carbonate (PC). Conductance histograms are shown in Figure 3.2a,b. We observe a larger conductance decay in TCB with $\beta = 0.14 \pm 0.03$ Å^{-1} ($\beta = 0.33 \pm 0.08$ unit^{-1}) and an increase in conductance with increasing length in PC yielding negative $\beta = -0.08 \pm 0.01$ Å^{-1} ($\beta = -0.17 \pm 0.03$ unit^{-1}) (Figure 3.2c). While a solvent-dependent β has been reported previously,[124] β for most molecules remains fairly constant with solvent. For example, measurements of oligophenylene diamines in TCB and PC yielded roughly the same conductance decay as in 1-nonanol.[94] We note that the larger β in TCB is still smaller than that for polyene. Furthermore, to the best of our knowledge, the negative β observed in PC is the first report of conductance increasing with the wire length.

The changes in conductance with solvent for cyanine wires could be due to two factors. First, solvents can induce shifts in energy level alignment between the molecular orbitals and the metal Fermi level, which can in turn affect β.[125] This is because solvent and target molecules can bind to under-coordinated gold atoms on the electrodes. The density and magnitude of surface dipoles induced at the metal/molecule interface due to these adsorbants can affect the Au work function, thereby altering level alignment for the molecular junction.[126] Second, solvatochromic effects are known for cyanine dyes. Shifts in the absorption spectrum of cyanine compounds have been reported to depend on the solvent[127] and on the counter ion.[128] However, it should be noted that these shifts are more significant for asymmetric merocyanine compounds. To explore the solvatochromism of **D1-D3**, we performed UV-Vis absorption measurements in TCB, 1-nonanol,

and PC.[94] We find the optical gap for **D1-D3** in TCB is about 0.05 eV smaller than compared with PC and 1-nonanol, which both display similar optical gaps. Such small changes in the optical gap are unlikely to alter trends in the off-resonant conductance measured here and thus cannot explain the solvent-dependent β.

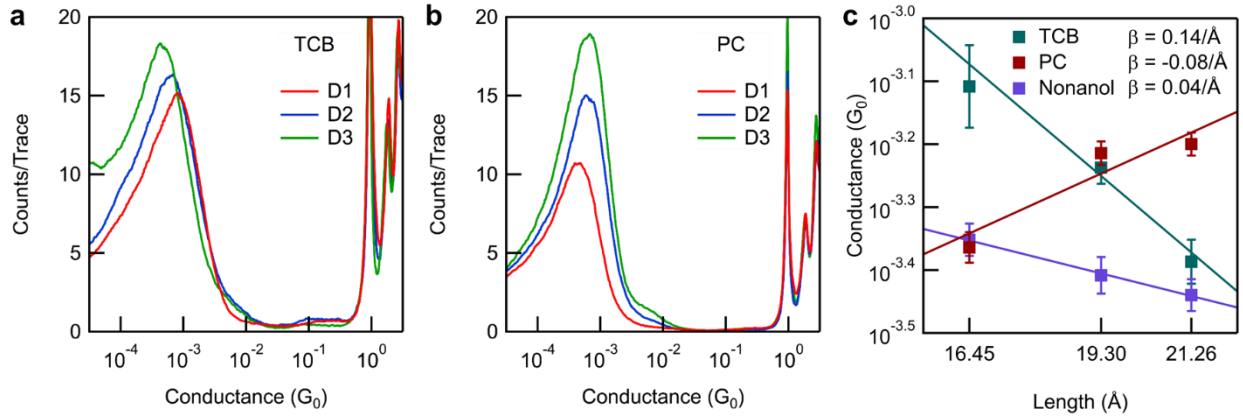

Figure 3.2. Solvent dependence of the cyanine dyes **D1-D3**. Conductance histograms of 10,000 traces at 90 mV bias of 0.1 mM **D1-D3** in (a) TCB and (b) PC. Histograms have 100 bins/decade. (c) Plot comparing the peak conductance values against molecular length in the three solvents. Solid straight lines indicate least-square fits to the data. Error bars represent distribution in peak values for 1,000 traces. A significant change in the conductance trends can be observed.

To provide additional insight into the electronic transport characteristics observed experimentally, we carried out *ab initio* quantum transport calculations based on DFT in model junctions for **D1-D3** attached to gold electrodes in an atop configuration (sulfur coordinated to a single gold atom, Figure 3.1c). For our simulations, we employed the AITRANSS module[129-130] interfaced with the FHI-aims package.[131] AITRANSS implements the non-equilibrium Green's function formalism applied to finite clusters with a simplified model for the reservoir self-energy.[94] We show in Figure 3.3a the calculated transmission spectra for **D1-D3.** The overall shape of the spectra for the three molecules is similar: the Fermi energy is situated inside the molecular HOMO-LUMO gap, closer to the HOMO-derived resonance. This agrees with bias-dependent experimental measurements we performed,[94] which indicate that the molecules are HOMO

conducting.[132] We find the theoretical zero-bias conductance to be nearly independent of the junction length, and close to 5.5×10^{-3} G_0. We note that the theoretical conductance values exceed the experimental ones by almost an order of magnitude. This discrepancy is due to approximations inherent in the exchange-correlation functionals used for the DFT-based transport calculations.[133-135] The calculated conductance attenuation factor is $\beta = 0.004 \pm 0.002$ Å^{-1} making these systems highly conductive and qualitatively different from the polyene case (Figure 3.1e). This small β is in excellent agreement with the measured results. Furthermore, small changes in level alignment can alter β from a small positive to a small negative value as shown in Figure 3.3a in agreement with the experimentally observed solvent-dependent β.

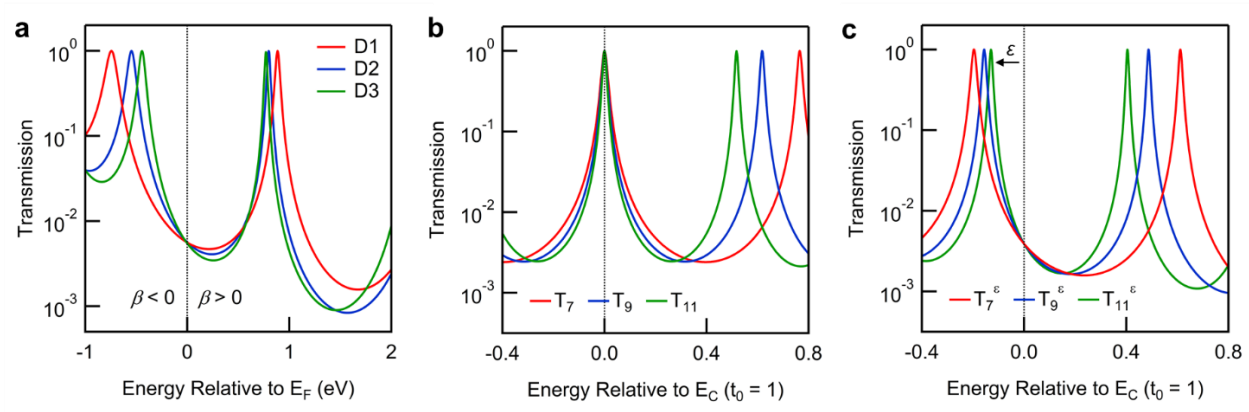

Figure 3.3. Conductance of cyanine dyes **D1-D3** compared to other oligomers. (a) DFT-based transmission spectra for **D1-D3** molecular junctions. The transmission decreases with the molecular length above E_F, whereas it increases with the system length below E_F. The conductance decay is therefore critically dependent on the alignment of the levels relative to E_F in agreement with the experimental results. (b),(c) Transmission spectra from a simplified Hückel model corresponding to the central backbone of **D1-D3** with (b) $t_1 = t_2$, $\Gamma = 0.05\ t_0$ and $\varepsilon = 0$ and (c) $t_1 = t_2$, $\Gamma = 0.05\ t_0$ and $\varepsilon = -0.4\ t_0$. In (c) ε shifts the resonances away from E_C (the on-site energy of carbon), but the length-independent transmission at E_F is maintained even in this off-resonance regime. The value of ε was chosen to model the trends from the DFT-based quantum transport calculations shown in (a).

To develop more detailed physical insights for these results we employ a simplified Su-Schrieffer-Heeger model[108, 110, 136] to model the conductance of a chain of N sp^2-hybridized carbon atoms. The on-site energy of the carbon atoms (E_C) is set to zero and alternating hopping along the chain is provided by $t_1 = t_0 e^{\delta}$ and $t_2 = t_0 e^{-\delta}$. When $\delta < 0$, $t_1 < t_2$ and t_1 and t_2 represent the nominal single and double bond couplings respectively, whereas, when $\delta > 0$, $t_1 > t_2$ and the situation is reversed. In this way, δ parameterizes the bond order alternation along the chain. To model the conductance, the terminal sites, $|1\rangle$ and $|N\rangle$ are coupled to metallic leads through constant leakage rates $\Gamma_{L(R)}$, yielding a Hamiltonian for the open system,

$$H_N = -\frac{i\Gamma_L}{2}|1\rangle\langle 1| - \frac{i\Gamma_R}{2}|N\rangle\langle N| + \sum_{k=1}^{N-1} t_0 e^{(-1)^k\delta}|k\rangle\langle k+1| + H.c. \tag{3.1}$$

Within the Landauer formalism, the conductance is given by $G = T(E_F)G_0$, where $T(E)$ is the transmission function, and E_F is the Fermi energy of the electrodes.[137] $T_N(E)$ for H_N, calculated using the single particle Green's function approach,[94] yields a simple and new closed-form expression at $E = E_C$,

$$T_N(E_C) = \begin{cases} \operatorname{sech}^2\left((N-1)\delta + \frac{1}{2}\ln\left(\frac{\Gamma_L}{2t_0}\right) + \frac{1}{2}\ln\left(\frac{\Gamma_R}{2t_0}\right)\right) & N = 2n \\ \operatorname{sech}^2\left((N-1)\delta + \frac{1}{2}\ln\left(\frac{\Gamma_L}{2t_0}\right) - \frac{1}{2}\ln\left(\frac{\Gamma_R}{2t_0}\right)\right) & N = 2n+1 \end{cases} \tag{3.2}$$

where the transmission takes on different forms for even and odd N. When $E_F \approx E_C$ (or equivalently when E_F is in the middle of the HOMO-LUMO gap) these expressions provide a qualitative description of the low bias conductance for this model. Importantly, we see that the length dependence for both even and odd chains is controlled by δ. For odd chains, when $t_1 = t_2$ or $\delta = 0$, coupling along the chain is uniform, and we find that transmission at E_F is resonant and does not depend on the system length, N, as shown in Figure 3.3b for chains of lengths corresponding to

D1-D3. This simplified model is consistent with the idea proposed earlier by Reimers and Hush[122] that systems with an odd number of C atoms will have low bond alternation and will exhibit smaller decays in conductance with length.

To model the cyanine series, we modify H_{2n+1} to include an on-site energy ε on the terminal sites (Eq. 3.3) to incorporate the effects of the nitrogen atoms on the chain.

$$H_{2n+1}^{\varepsilon} = \varepsilon(|1\rangle\langle 1| + |2n+1\rangle\langle 2n+1|) + H_{2n+1} \tag{3.3}$$

If we consider a symmetrically coupled junction ($\Gamma_L = \Gamma_R \equiv \Gamma$), the transmission at $E = E_C$ becomes:

$$T_{2n+1}^{\varepsilon}(E_C) = \frac{\Gamma^2}{4\varepsilon^2 + \Gamma^2} T_{2n+1}(E_C). \tag{3.4}$$

We can see that the altered on-site energy of the terminal sites modifies $T_{2n+1}(E_C)$ by a factor that is analogous the inverse of a "contact resistance" which, as expected, does not affect the transmission trends with the chain length. The transmission spectra for chains of lengths corresponding to **D1-D3** are plotted for $\delta = 0$ in Figure 3.3c. The inclusion of the more electronegative nitrogen atoms in cyanine ($\varepsilon < 0$) shifts the resonance below E_F. This "contact resistance" reduces the transmission value at E_F; however, the length dependent behavior of the transmission remains the same. Namely, transmission remains length independent when $\delta = 0$. We find qualitative agreement between the DFT based transmissions shown in Figure 3.3a and those from the Hückel model shown Figure 3.3c choosing t_0 to be approximately 2 eV. Therefore, we can conclude that the low conductance attenuation observed for these odd-numbered chains **D1-D3** is related to the small bond order alternation present in cyanine dyes, which is described in this simplified model by $\delta \approx 0$. This is consistent with the molecular geometries of the DFT-optimized structures which show that the bond length alternation for **D1-D3** is significantly smaller than a conventional polyene.[94]

The length dependence of the transmission of Eq. 3.2 can be further characterized by defining $\beta \equiv -\partial/\partial n \ln(T)$; if β is independent of n, we have $T \sim e^{-\beta n}$. For our model, assuming a length-independent δ, we obtain a closed-form expression for β at $E = E_C = E_F$:

$$\beta_N = \begin{cases} 4\delta \tanh\left((N-1)\delta + \frac{1}{2}\ln\left(\frac{\Gamma_L}{2t_0}\right) + \frac{1}{2}\ln\left(\frac{\Gamma_R}{2t_0}\right)\right) & N = 2n \\ 4\delta \tanh\left((N-1)\delta + \frac{1}{2}\ln\left(\frac{\Gamma_L}{2t_0}\right) - \frac{1}{2}\ln\left(\frac{\Gamma_R}{2t_0}\right)\right) & N = 2n+1 \end{cases} \tag{3.5}$$

In the experimentally relevant regime of poor coupling ($\Gamma_L \approx \Gamma_R << t_0$), small N and small $|\delta|$, β_N can be simplified to yield[108]:

$$\beta_N \approx \begin{cases} -4\delta & N = 2n \\ 4(N-1)\delta^2 & N = 2n+1 \end{cases} \tag{3.6}$$

As discussed above, we see that δ is the critical parameter that controls the conductance decay for polyenes and polymethines, respectively. In agreement with previous studies, the large conductance attenuation for polyenes (positive β_{2n}) can be related to its highly alternant structure (negative δ).[108] Interestingly when δ is positive, polyene β_{2n} is negative and conductance should increase with length for small N. From a valence bond perspective, this regime corresponds to polyenes with diradical character, where the terminal bonds are single bonds as opposed to double bonds ($t_1 > t_2$). In the long-chain polyacetylene limit ($N \rightarrow \infty$), β for even and odd chains converge to the same value: $\beta_\infty = 4|\delta|$. This expression for β_∞ can also be derived from the complex band structure for an infinite polyene wire, i.e., polyacetylene.[94]

In conclusion, we have experimentally observed that the conductance attenuation with length for cyanines, a class of polymethine compounds, is significantly lower than for polyenes. The β value for this cyanine series represents one of the smallest β values ever reported in an off-resonant transport regime. This result is supported by first-principle calculations. Furthermore, we

develop a simplified model that suggests that the low β measured is not an anomalous result but rather a general property of short chain polymethine systems. We show that polymethine systems are expected to exhibit smaller conductance attenuation than polyenes because of the smaller degree of bond order alternation. However, we note that low bond order alternation is a short chain effect; we would expect β to increase in the polymeric limit resembling a polyene. In addition, we show that we are able to control the sign of β (and therefore the conductance trends with the molecular wire length) through a choice of solvent. Notably, in PC, we measure a negative β value, where we demonstrate a conductance increases with increasing molecular length for the first time. These novel electronic properties of polymethine compounds can be exploited in future molecular electronic components.

Chapter 4: Visualizing quantum interference

4.1 Preface

This chapter is adapted from the manuscript entitled *Visualizing Quantum Interference in Molecular Junctions* by Suman Gunasekaran, Julia Edith Greenwald, and Latha Venkataraman.[138] Discussions with Daniel Hernangómez-Pérez and Ferdinand Evers are gratefully acknowledged.

4.2 Abstract

Electron transport across a molecular junction is characterized by an energy-dependent transmission function. The transmission function accounts for electrons tunneling through multiple molecular orbitals (MOs) with different phases, which gives rise to quantum interference (QI) effects. Since the transmission function comprises both interfering and noninterfering effects, individual interferences between MOs cannot be deduced from the transmission function directly. Herein, we demonstrate how the transmission function can be deconstructed into its constituent interfering and noninterfering contributions for any model molecular junction. These contributions are arranged in a matrix and displayed pictorially as a QI map, which allows one to easily identify individual QI effects. Importantly, we show that exponential conductance decay with increasing oligomer length is primarily due to an increase in destructive QI. With an ability to "see" QI effects using the QI map, we find that QI is vital to all molecular-scale electron transport.

4.3 Introduction

Quantum mechanics dictates electron transport at the nanoscale.[11, 139-141] In single-molecule junctions, where the device size is comparable to the electronic phase coherence length, the phase of the tunneling electron wave influences transport characteristics.[142] Phase differences arise when an electron wave traverses multiple conduction *pathways*, which introduces quantum interference (QI) effects that may either increase (constructive QI) or decrease (destructive QI) the probability

of electron tunneling.[90, 143-147] *Pathways* can be either atomic orbital-derived spatial paths or molecular orbital (MO)-derived transmission channels. MO-based QI is more generally applicable to molecular junctions and will be the focus of the discussion herein.

The theoretical description of QI between MOs has been well-established.[140, 148] An electron incident on a molecular junction may tunnel via multiple MOs. The probability of tunneling via a single level, the ith MO, is described by a Lorentzian transmission function, denoted here as $T_i(E)$,

$$T_i(E) = \frac{\gamma_i^2}{(E - \varepsilon_i)^2 + \gamma_i^2} \tag{4.1}$$

where E is the energy of the incident electron, ε_i is the peak of the Lorentzian and corresponds to the energy level of the ith MO, and γ_i denotes the strength of the coupling between the MO and the electrodes.[149-150] An electron wave transmitted through the level experiences a phase shift relative to the incident wave (Figure 4.1).[151-152] The amplitude and phase of the transmitted wave are described by a complex transmission coefficient t_i, where the transmission probability is the modulus squared of the transmission coefficient,[153-154]

$$T_i = |t_i|^2 \tag{4.2}$$

For a molecular junction, the total transmission probability is the modulus squared of the sum of the complex transmission coefficients for all MOs. For a molecule with two MOs, i and j, the total transmission probability, $T_{i+j}(E)$, is,

$$T_{i+j} = \left|t_i + t_j\right|^2 = |t_i|^2 + \left|t_j\right|^2 + t_i t_j^* + t_i^* t_j \tag{4.3}$$

The total transmission probability is not merely the sum of the individual transmission probabilities for each MO ($T_{i+j} \neq |t_i|^2 + \left|t_j\right|^2$). The cross-terms in Eq. 4.3 represent QI between the ith and j^{th} MO, which we denote $T_{ij}(E)$,[155]

$$T_{ij} = t_i t_j^* + t_i^* t_j = 2\sqrt{T_i T_j}\cos(\theta_i - \theta_j) \tag{4.4}$$

where θ_i and θ_j are the complex phases of t_i and t_j, respectively. If θ_i equals θ_j, there is constructive interference between the ith and j^{th} MO, whereas if θ_i and θ_j differ by 180° there is destructive interference. Hence, QI can be understood in terms of phase differences between MO-derived transmission coefficients.[146]

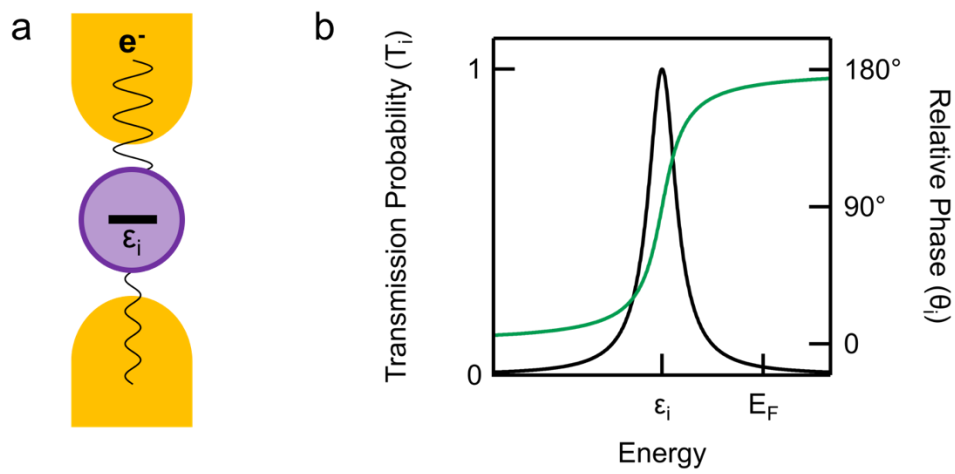

Figure 4.1. a) Schematic of molecular junction with a single level, i. The amplitude and phase of the transmitted wave differ from those of the incident wave. b) Tunneling probability (black) and phase of the transmission coefficient (green) as a function of energy for transport across a single level with energy ε_i.

Yoshizawa and coworkers have developed a set of rules to predict the presence of constructive or destructive QI near the Fermi energy for certain simple situations.[44] These rules rely on determining the relative phases of the transmission coefficients of the frontier MOs. Others have developed various pen-and-paper methods for predicting the existence of anti-resonances at the Fermi energy.[11, 147, 156] However, QI is relevant to all molecular junctions, not simply those with transmission functions featuring anti-resonances. A general, quantitative technique to easily calculate and visualize QI in all systems is currently lacking. As a result, QI is often overlooked in transmission functions where anti-resonances do not appear.

4.4 Results and discussion

To model electron transport in a molecular junction, the Green's function approach is commonly employed, for which the transmission function is calculated as,[157]

$$T(E) = \mathrm{Tr}[\Gamma_L G \Gamma_R G^\dagger] \tag{4.5}$$

where G and $G^\dagger$ are the retarded and advanced Green's functions, and Γ_L and Γ_R are matrices describing the coupling to the left and right electrodes, respectively. The Green's function is typically expressed in the atomic basis.[158] To extract QI between MOs, a change of basis must be performed to transform the Green's function matrix from the atomic basis to the MO basis. This transformation can be achieved using a change of basis matrix, P, whose columns comprise the eigenvectors of G and for which the matrix $P^{-1}GP$ is diagonal. In the MO basis, the transmission function, T, becomes a sum over all MOs, encompassing both the noninterfering, T_i, and interfering terms, T_{ij},[140, 159]

$$T = \sum_i T_i + \sum_{i>j} T_{ij} \tag{4.6}$$

To provide a simple way of calculating the noninterfering and interfering terms, we introduce a matrix, Q, defined as,

$$Q(E) = (P^\dagger \Gamma_L G P) \circ (P^{-1} \Gamma_R G^\dagger P^{-1\dagger})^{\mathrm{T}} \tag{4.7}$$

where $(\circ)$ denotes the entrywise product and $(\cdot)^{\mathrm{T}}$ denotes the matrix transpose.[138] The noninterfering transmission probability for the ith MO is simply given by the ith diagonal element of Q,

$$T_i = Q_{ii} \tag{4.8}$$

and the QI between the ith and j^{th} MOs is given by the sum of the $(i,j)^{th}$ and $(j,i)^{th}$ elements of Q,

$$T_{ij} = Q_{ij} + Q_{ji} \tag{4.9}$$

Note that since Q is Hermitian ($Q_{ji}^{*} = Q_{ij}$), both T_i and T_{ij} are real. Additionally, one can conveniently calculate total transmission excluding interference contributions, denoted here as $\hat{T}$, from the trace of Q,

$$\hat{T} = \sum_{i} T_i = \mathrm{Tr}[Q] \tag{4.10}$$

The Q matrix therefore deconstructs the transmission function into its constituent interfering and noninterfering components. Accordingly, a summation of all the elements in Q recovers the total transmission function,

$$T = \sum_{ij} Q_{ij} \tag{4.11}$$

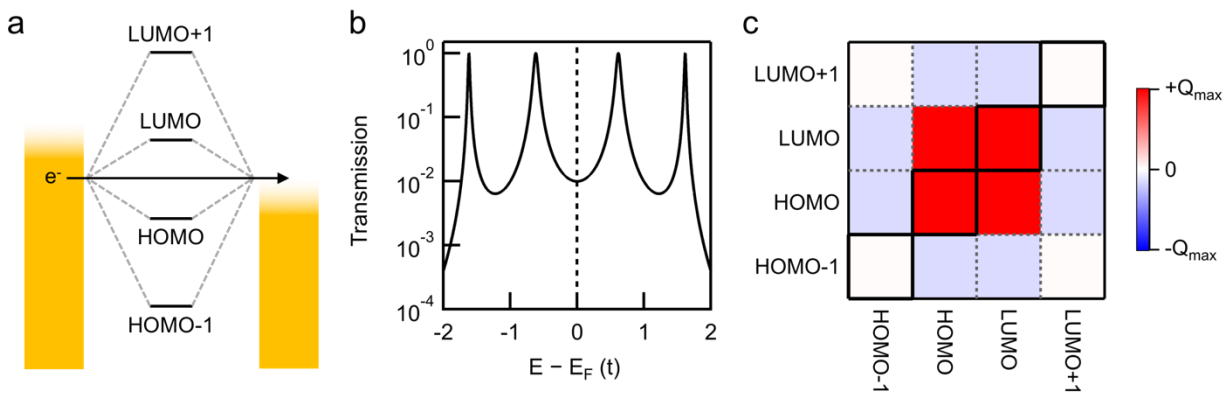

Figure 4.2. a) Schematic of MO interference for butadiene. b) Transmission function for Hückel model of 1-4 butadiene. Energy is expressed in units of t. The dashed line highlights E_F. c) QI map for 1-4 butadiene at E_F. The main diagonal is highlighted in bold. The off-diagonal squares represent constructive (red) and destructive (blue) QI between MOs. The color scale is normalized to $Q_{max} = \max|Q_{ij}|$.

To demonstrate the utility of the Q matrix in elucidating QI effects, we consider a butadiene molecular junction connected to the electrodes at the 1 and 4 positions (i.e., 1-4 butadiene) as an initial example (Figure 4.2). The π-system of butadiene has four MOs—HOMO-1, HOMO, LUMO, and LUMO+1—which can be modeled at the Hückel level with hopping integral t.

Although a tunneling electron "sees" all four MOs as it transits the molecule, which gives rise to QI (Figure 4.2a),[146] there are no discernable QI effects in the transmission function (Figure 4.2b). Specifically, the prominence of the four Lorentzian resonance peaks leads one to naïvely believe that the transmission function is merely the sum of the four noninterfering MO-derived resonances. However, calculation of the Q matrix (Eq. 4.7) elucidates that QI is in fact important at all energies and between all orbitals.

The Q matrix for 1-4 butadiene at the Fermi energy (E_F) is presented in the form of a heat map (Figure 4.2c), henceforth referred to as a QI map. Since butadiene has four π-based MOs, the QI map is a four-by-four grid. Each square in row i and column j corresponds to the real part of element Q_{ij}. The color of each square represents the sign of the matrix element, with red representing positive values and blue representing negative values. Note that the QI map depicts the Q matrix flipped relative to standard convention (e.g., (1,1) is in the lower left corner). The squares along the main diagonal (highlighted in bold) represent the non-interfering transmission terms (e.g., Q_{ii}) and will necessarily be red. The off-diagonal squares may either be red (constructive QI) or blue (destructive QI). The QI map is always symmetric about the main diagonal since Q is Hermitian. The QI map for 1-4 butadiene shows that in addition to individual contributions from the HOMO and the LUMO, there is constructive interference between the HOMO and LUMO at E_F. Interestingly, the magnitude of the constructive interference between the HOMO and LUMO is the comparable to the individual HOMO and LUMO contributions. The HOMO and LUMO also destructively interfere with the HOMO-1 and LUMO+1 (light blue regions in Figure 4.2c). The QI in 1-4 butadiene originates in part from the phase of the MO coefficients at the sites of contact with the electrodes.[44, 160] Contacting the molecule at different sites, i.e., 1-2, 1-3, and 2-3 butadiene, yields vastly different QI maps.[138, 161]

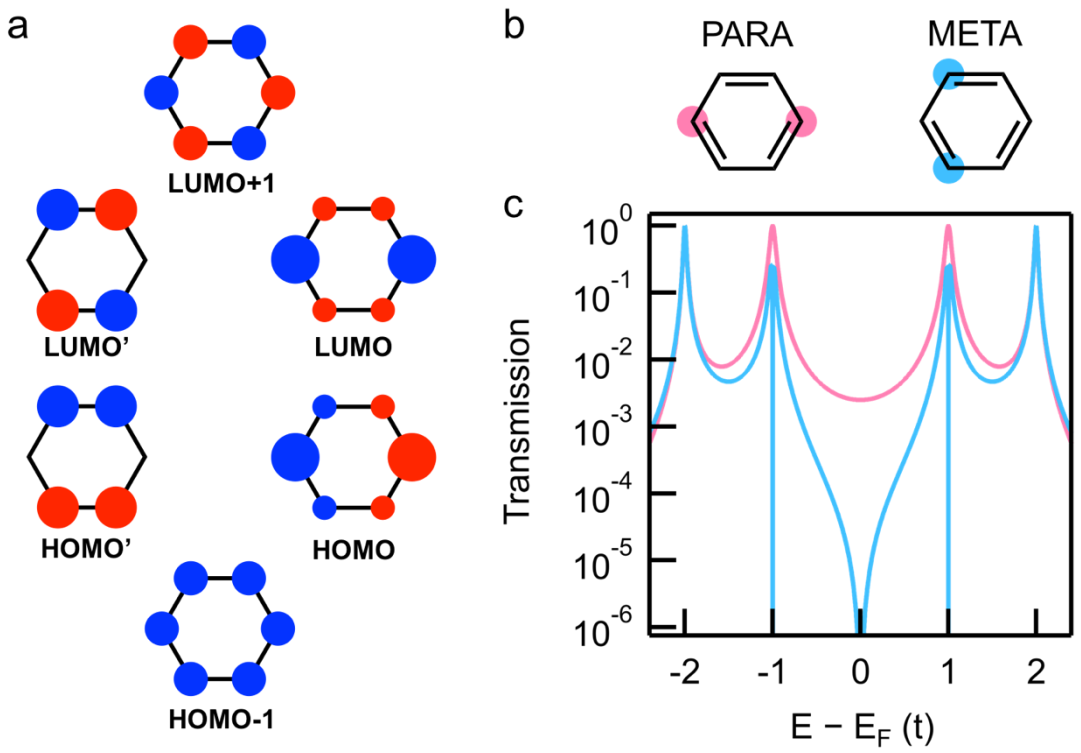

Figure 4.3. a) π-based MOs of benzene. b) Electrical contact sites for molecular junctions of para- and meta-benzene. c) Transmission functions for para- (pink) and meta-benzene (blue).

Comparing meta- and para-benzene further illustrates the connection between QI and MO coefficients. In isolation, benzene possesses doubly degenerate frontier MOs (Figure 4.3a). When coupled to the electrodes, this degeneracy is lifted, such that the HOMO and HOMO' and the LUMO and LUMO' split in energy slightly. Furthermore, the energetic ordering of these MOs depends on the electrical contact sites. For para-benzene, HOMO and LUMO are the frontier MOs, whereas for meta-benzene, HOMO' and LUMO' are the frontier MOs. The sites of electrical contact for each molecular junction are highlighted in Figure 4.3b. The contact sites are explicitly positioned to correspond with the degenerate MOs of the isolated molecule presented in Figure 4.3a and can be justified by orbital symmetry arguments.[162] The difference in contact sites causes meta- and para-benzene molecular junctions to exhibit starkly different transmission functions (Figure 4.3c).[143, 145] Para-benzene exhibits constructive interference at E_F, whereas meta-benzene

exhibits complete destructive interference at E_F. At the frontier orbital energies (± 1 t in Figure 4.3c), para-benzene has a resonance feature with unit transmission while meta-benzene has an anti-resonance.

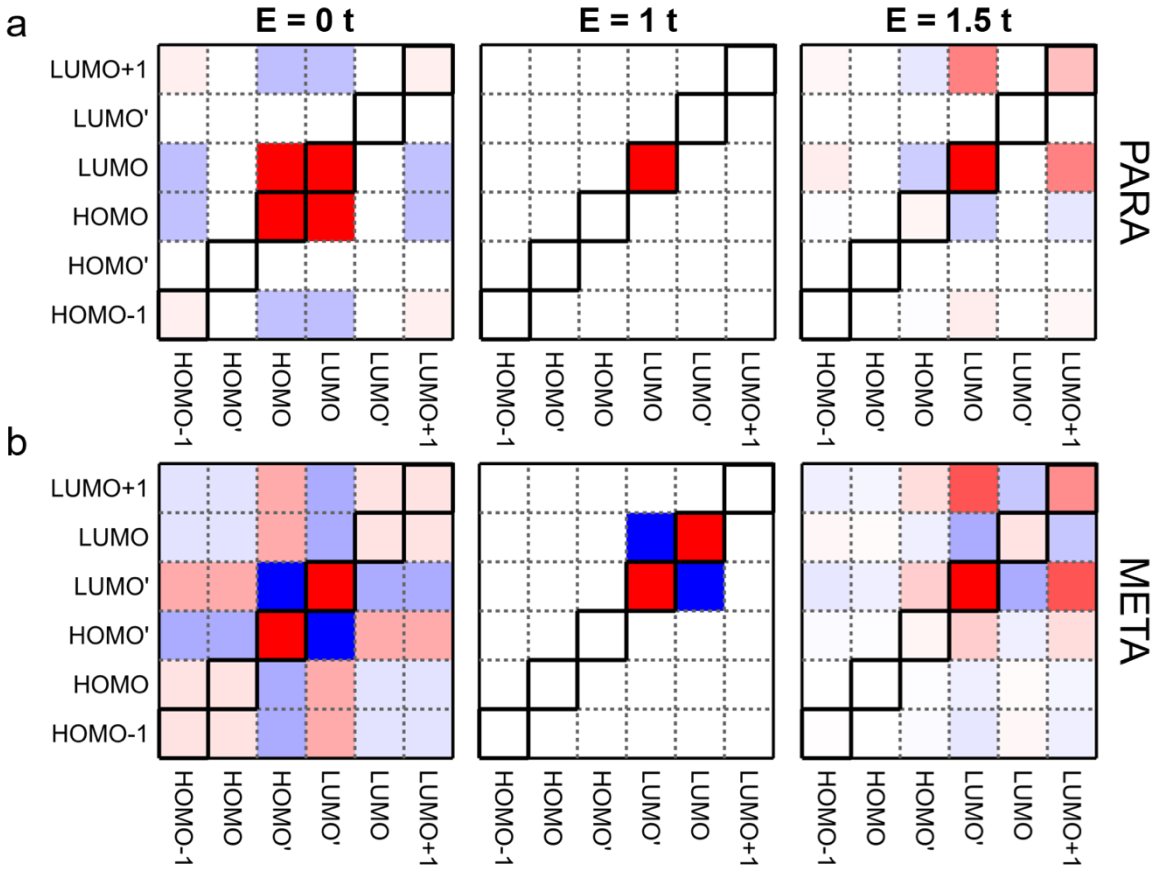

Figure 4.4. QI maps for a) para- and b) meta-benzene at three different energies: 0 t (left); 1 t, at LUMO resonance (center); and 1.5 t, between LUMO and LUMO+1 (right). Color scale for each heat map is normalized to Q_{max} and therefore is different for each heat map.

The QI maps for para- and meta-benzene illustrate the energy-dependent QI phenomena observed in the transmission functions (Figure 4.3c). In Figure 4.4, QI maps for para- and meta-benzene are presented at three different energies: E_F (E = 0 t), the energy of the LUMO resonance (E = +1 t), and an energy between the LUMO and LUMO+1 resonances (E = +1.5 t). In para-benzene, the HOMO and LUMO constructively interfere at E_F (Figure 4.4a, left). Since the MO

coefficients of HOMO' and LUMO' are zero at the electrical contact sites (Figure 4.3a,b), HOMO' and LUMO' do not contribute to transmission in para-benzene. This gives rise to the many white squares (i.e., where the *Q*-matrix is zero) in the QI maps, most visible in the QI map at $E = 0$ t. Although there are no signatures of destructive interference in the transmission function, the blue squares (Figure 4.4a, left) indicate that the HOMO and LUMO destructively interfere with the HOMO-1 and LUMO+1 akin to 1-4 butadiene (Figure 4.2c). The QI map for para-benzene at $E = +1$ t, indicates that transmission is dominated by the LUMO (Figure 4.4a, center). While other contributions to transmission exist at $E = +1$ t, they are insignificant compared to the contribution from the LUMO and are not visible with the chosen color scaling.

The QI maps for meta-benzene display strikingly different QI features than those of para-benzene. In meta-benzene, the destructive QI between HOMO' and LUMO' at E_F (Figure 4.4b, left) is equal to the noninterfering contributions of HOMO' and LUMO', which gives rise to the anti-resonance feature observed in the transmission function (Figure 4.3c). The QI map at $E = +1$ t shows strong destructive QI between the LUMO and LUMO' (Figure 4.4b, center), which accounts for the sharp spike in the transmission function (Figure 4.3c). The destructive interference is due to the opposite phases of the MO coefficients for LUMO and LUMO' at the contact sites. Beyond the LUMO resonance ($E = +1.5$ t), we observe some interesting changes in color in the QI maps for both para- and meta-benzene. In particular, the interference between the frontier MOs changes from constructive to destructive QI for para-benzene (Figure 4.4a, right), and changes from destructive to constructive QI for meta-benzene (Figure 4.4b, right). These changes in QI can be attributed to the 180° difference in the phase of the transmission coefficient on either side of a resonance (Figure 4.1b).[163]

While the above examples are illuminating from a theoretical standpoint, in practice, molecular junctions tend to comprise longer, oligomeric systems. To probe transport properties experimentally, oligomers of varying length are often studied. It has been observed that conductance decays exponentially with oligomer length,

$$G \sim e^{-\beta n} \tag{4.12}$$

where G is the conductance, n is the number of monomer units, and β is the fitting parameter characteristic of the molecular backbone.[95, 164] This exponential decay can be reproduced at the Hückel level of theory.[162, 165]

To explore the relationship between exponential decay and QI, we model the transmission functions for the oligophenyl series (Figure 4.5a, n = 1–6). Figure 4.5b and Figure 4.5c show the transmission functions for all six oligomers including QI (Eq. 4.5) and omitting QI (Eq. 4.10), respectively. In both cases, the conductance—approximated as the transmission at E_F—decreases as a function of length (Figure 4.5d). When QI is included, transmission decays exponentially; however, when QI is omitted, the decay with length is substantially reduced. The modest decrease in conductance observed when QI is omitted can be attributed to a decrease in coupling to the electrodes. As the number of monomer units increases, the MOs become delocalized over longer distances and orbital density throughout the molecular backbone is reduced, which weakens coupling between the MOs and the electrodes. As coupling decreases, the Lorenztian peaks become narrower and transmission at E_F decreases. Previously, decreased coupling has been considered the principle cause of exponential decay with length in molecular junctions.[166] However, direct comparison of conductance with and without QI (Figure 4.5d) clearly shows that the exponential decay is predominantly due to destructive QI. A similar trend is observed for the polyene series.[138]

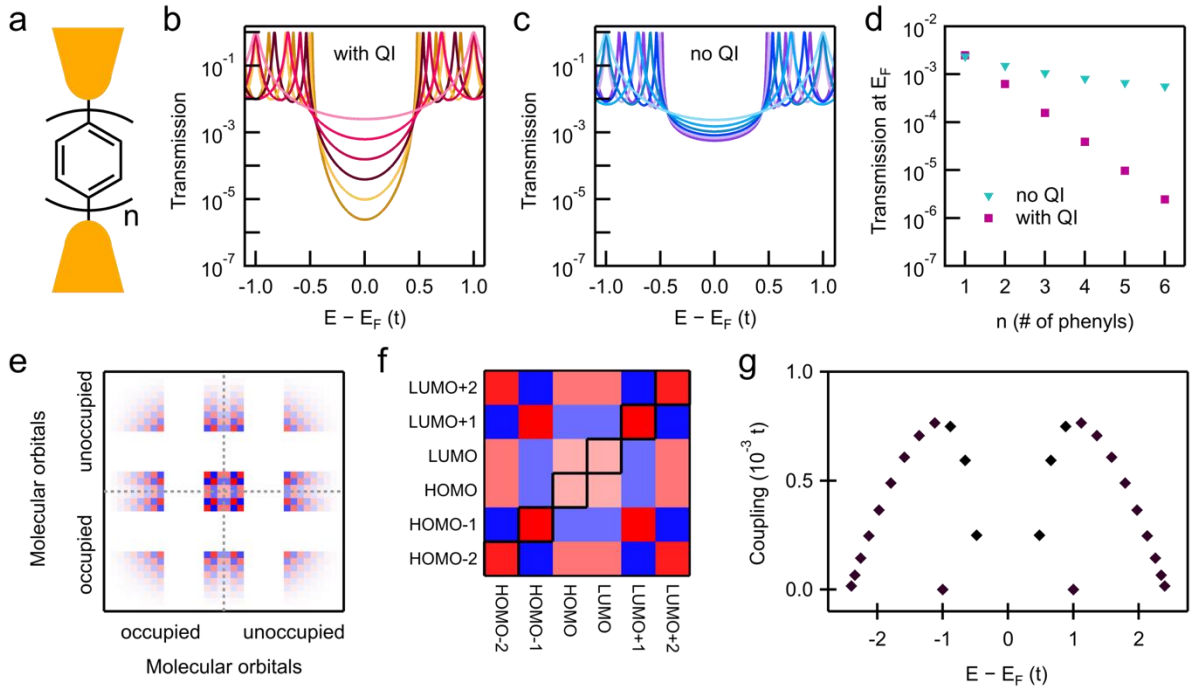

Figure 4.5. a) Schematic of oligophenyl molecular junction (n = 1–6). b,c) Transmission (b) with QI and (c) without QI for oligophenyl series. d) Transmission at E_F as a function of length for oligophenyl series with QI (magenta) and without QI (turquoise). e,f,g) Analysis of n = 6 oligophenyl: (e) QI map at E_F for all π-based MOs, dashed lines placed in between HOMO and LUMO, (f) Enlarged QI map at E_F for the six MOs closest to E_F, (g) MO-electrode coupling vs. MO energy for all MOs.

We next consider the QI map for n = 6 oligophenyl at E_F (Figure 4.5e) to understand the origin of the destructive QI. The striking pattern of white squares in the QI map comes from the presence of multiple degenerate uncoupled MOs, which are combinations of HOMO' and LUMO' MOs of para-benzene (Figure 4.4a). The enlarged QI map for the six MOs closest to E_F (Figure 4.5f) demonstrates that both the noninterfering and interfering transmission contributions from the HOMO-2, HOMO-1, LUMO+1 and LUMO+2 are greater than those from the HOMO and LUMO. Therefore, even for an oligomer of modest length, the conductance at Fermi is significantly influenced by MOs beyond the frontier MOs. In particular, there exists strong destructive QI between these non-frontier MOs.

Additional insight can be gleaned from the energy-coupling diagram (Figure 4.5g), in which MO-electrode coupling versus MO energy is plotted for all 36 π-MOs of n = 6 oligophenyl. This diagram is produced by calculating the real and imaginary parts of the eigenvalues of H+Σ, where H is the Hamiltonian of the molecule and Σ is the self-energy of the electrodes. The degenerate, uncoupled MOs that produce the white squares in the QI map accumulate as single points at $E = \pm 1$ t in the energy-coupling diagram. Importantly, the diagram shows that the HOMO-2, HOMO-1, LUMO+1, and LUMO+2 are better coupled than the HOMO and LUMO and are energetically very close to E_F. This explains why these non-frontier MOs contribute more significantly to transmission at E_F, as evidenced by the QI map (Figure 4.5f). The QI maps for n = 1–6 for both the polyene and oligophenyl series show that interfering contributions from non-frontier orbitals increase as a function of oligomer length.[138] Notably, both constructive and destructive QI increase with oligomer length, but destructive QI increases more. Therefore, exponential conductance decay with oligomer length can be attributed to an increase in destructive QI from non-frontier MOs that increasingly influence transport at E_F.

In conclusion, we have shown that quantum interference is ubiquitous in molecular-scale transport. The presence of multiple MOs in a molecular junction means that an incident electron wave may experience interference effects between MOs as it tunnels through the molecular junction. Since QI effects are hard to discern in standard transmission functions, we present a QI map to easily visualize distinct destructive and constructive interference effects between MOs. Using the QI map, we show that exponential decay in conductance as a function of length is due to increasing destructive QI from non-frontier MOs. Therefore, care must be taken when reducing a molecule to its frontier molecular orbitals and modeling transmission near the Fermi energy using a single Lorentzian or a sum of two Lorentzians. These results highlight that transport in a molecular junction must be viewed holistically, and transmission cannot simply be segregated into contributions from individual, noninterfering MOs.

Chapter 5: Single-electron currents in single-cluster devices

5.1 Preface

This chapter is adapted from the manuscript entitled *Single-electron Currents in Designer Single-Cluster Devices* by Suman Gunasekaran, Douglas Reed, Daniel Paley, Amymarie Bartholomew, Latha Venkataraman, Michael Steigerwald, Xavier Roy, and Colin Nuckolls.[167] Valuable discussions with E-Dean Fung and Louis E. Brus are gratefully acknowledged.

5.2 Abstract

Atomically-precise clusters can be used to create single-electron devices wherein a single redox-active cluster is connected to two macroscopic electrodes via anchoring ligands. Unlike single-electron devices comprising nanocrystals, these cluster-based devices can be fabricated with atomic precision. This affords an unprecedented level of control over the device properties. Herein, we design a series of cobalt chalcogenide clusters with varying ligand geometries and core nuclearities to control their current-voltage (I-V) characteristics in a scanning tunneling microscope-based break junction (STM-BJ) device. First, the device geometry is modified by precisely positioning junction-anchoring ligands on the surface of the cluster. We show that the I-V characteristics are independent of ligand placement, confirming a sequential, single-electron tunneling mechanism. Next, we chemically fuse two clusters to realize a larger cluster dimer that behaves as a single electronic unit, possessing a smaller reorganization energy and more accessible redox states than the monomeric analogues. As a result, dimer-based devices exhibit significantly higher currents and can even be pushed to current saturation at high bias. Owing to these controllable properties, single-cluster junctions serve as an excellent platform for exploring incoherent charge transport processes at the nanoscale. With this understanding, as well as

properties such as nonlinear I-V characteristics and rectification, these molecular clusters may function as conductive inorganic nodes in new devices and materials.

5.3 Introduction

A prototypical single-electron device comprises a nanoscale quantum dot isolated from the leads by tunnel junctions.[168-170] Current is driven by sequentially charging and discharging the quantum dot one electron at a time. Single-electron devices can serve as nanoscale circuit elements such as transistors and switches[171-172] and can be incorporated into novel quantum devices that require single-electron manipulations.[173-176] Conventional single-electron devices are fabricated lithographically; this limits both the miniaturization and the chemical uniformity of the quantum dots. Since single-electron devices require the energy level spacing of the quantum dot to be larger than the thermal energy ($\Delta E \gg k_B T$), conventional single-electron devices can only operate at low temperatures.[176] Room temperature operation requires quantum dot sizes below ten nanometers to achieve sufficient energy level spacing.[177-178] This can be achieved with scanning probe patterning[179-180] or through the use of nanocrystals.[178, 181-184] However, these approaches lack the requisite control over the properties of the quantum dot and the tunnel junctions needed for reproducible operation across several devices.

Single-molecule circuits can overcome these limitations since the quantum dot and the tunnel junctions can be defined with atomic precision using synthetic chemistry.[26, 139, 185-186] Although single-electron transfer processes at the molecular level have recently garnered significant interest,[187-189] room temperature demonstrations of single-electron devices using single molecules are limited.[190-193] The reason for this dearth is that single-molecule circuits typically comprise organic molecules or organometallic complexes[194] whose transport is dominated by coherent tunneling pathways.[144] As a result, even for redox-active molecules,[195-197]

charging/discharging processes are overwhelmed by through-bond tunneling across the molecular junction.[198-201]

In contrast, redox-active atomically-precise inorganic clusters[202-203] do not suffer from the limitations of relatively simple organic molecules. In a single-cluster junction, the electronic states of the cluster are localized on the cluster and have limited spatial overlap with the leads.[204] As a result, the probability of tunneling coherently across the entire junction is negligible. Accordingly, previous work on hexanuclear cobalt sulfide clusters demonstrated that, at low biases, significant current suppression is observed due to the lack of coherent tunneling.[204] To drive a current, higher biases must be applied in order to energetically enable charging/discharging of the cluster.[205] In this work, we elucidate the mechanism of current flow through inorganic clusters by synthesizing a series of cobalt selenide clusters of varying core sizes and surface ligand distributions. We performed room temperature current-voltage (I-V) measurements on thousands of single-cluster devices using the scanning tunneling microscope-based break junction (STM-BJ) technique.[21] Our results firmly establish that the mechanism of charge transport is through a sequential tunneling process. The I-V characteristics obey semi-classical heterogeneous electron transfer theory where electron transfer rates are influenced by reorganization energy. Due to their atomic precision and synthetic diversity,[206-207] single-cluster junctions are ideally suited to explore single-electron currents at the molecular scale towards realizing functional devices.

5.4 Results and discussion

In the first series of experiments, we investigate the role of device geometry on the transport properties. The clusters [*trans*-$Co_6Se_8(PEt_2(p\text{-}C_6H_4SMe))_2(PEt_3)_4][PF_6]$ (**trans-L2**), [*cis*-$Co_6Se_8(PEt_2(p\text{-}C_6H_4SMe))_2(PEt_3)_4][PF_6]$ (**cis-L2**), and $[Co_6Se_8(PEt_2(p\text{-}C_6H_4SMe))_6][BF_4]$ (**L6**) all contain identical $[Co_6Se_8]^{1+}$ cores but differ in the number and placement of junction-anchoring

$PEt_2(p\text{-}C_6H_4SMe)$ ligands (Figure 5.1a-c). These anchoring ligands allow the clusters to bind to gold electrodes through thioether linkages. Although the cluster features selenium atoms, previous studies demonstrated that junction formation does not occur without these thioether groups, confirming the necessity of these anchoring ligands.[208] The **trans-L2** and **cis-L2** clusters exclusively enable the formation of junctions in either the *trans* or *cis* configurations, respectively. In **L6**, the cluster can bind in multiple possible configurations. By synthetically controlling the ligand positions, we can precisely engineer the geometry of the single-cluster device and probe how it affects the transport behavior.

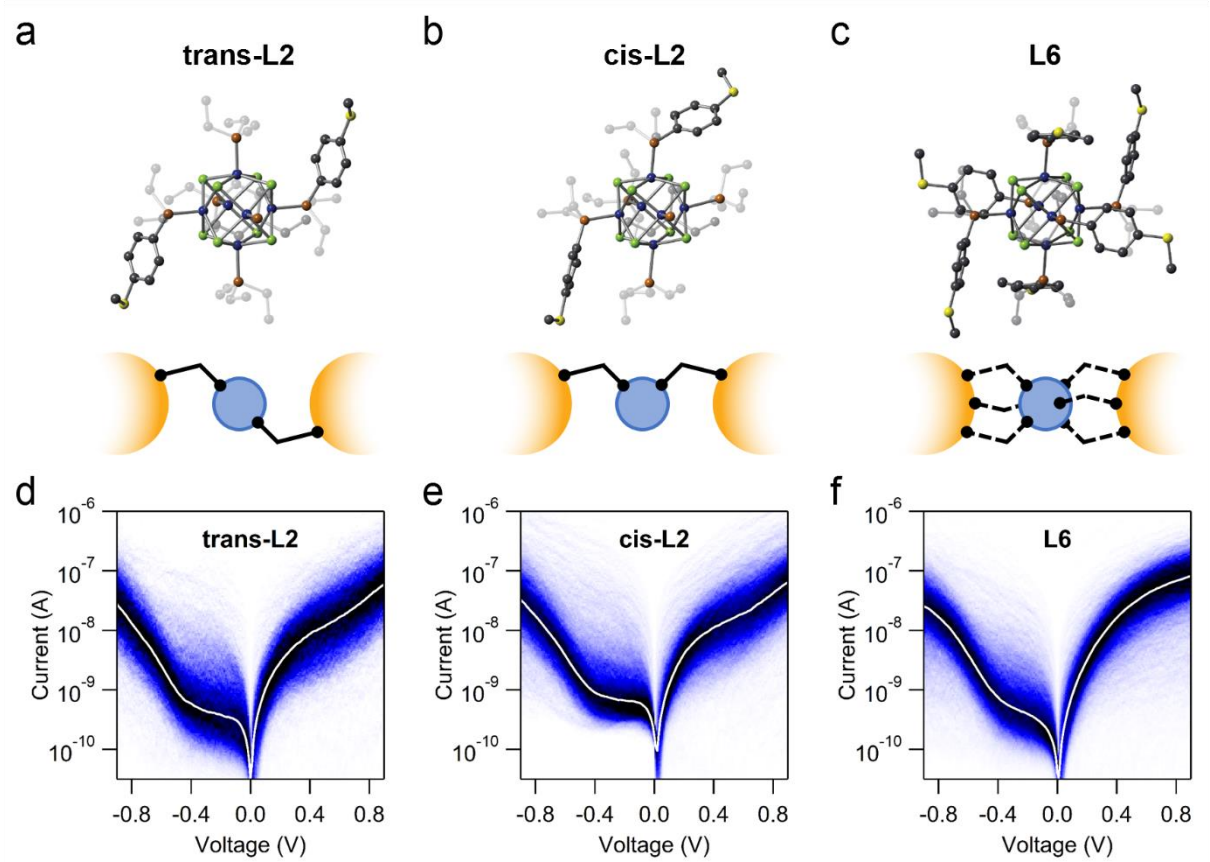

Figure 5.1. (a,b,c) Solid state structures of (a) **trans-L2**, (b) **cis-L2**, and (c) **L6**, as determined from SCXRD. Grey, blue, orange, yellow, and green spheres represent C, Co, P, S, and Se, respectively; ethyl groups are faded and H atoms are omitted for clarity. Schematic molecular junctions are also presented highlighting the differing connectivities between the cluster (blue) and the electrodes (gold). The variable connectivity of **L6** is indicated with dashed lines. (d,e,f) 2D absolute current versus voltage histograms compiled from several thousand traces for (d) **trans-L2** (2,144 traces),

(e) **cis-L2** (3,585 traces), and (f) **L6** (4,208 traces). The current is logarithmically binned (40 bins/decade) and the voltage is linearly binned (100 bins/V). The average I-V curves (white) were obtained by fitting each vertical slice of the 2D I-V histograms with a Gaussian and plotting the peak position against voltage.

While the synthesis and study of **L6** with six identical surface ligands have been reported previously,[208] the preparation of **trans-L2** and **cis-L2** is only made possible by recent synthetic developments that yield substitutionally diverse yet precise clusters.[209] We synthesized and isolated, in high yields, clusters with positionally substituted carbon monoxide ligands, *trans*-$Co_6Se_8(CO)_2(PEt_3)_4$ and *cis*-$Co_6Se_8(CO)_2(PEt_3)_4$. These carbonyls are photolytically labile, and irradiation of mixtures of these precursors and $PEt_2(p\text{-}C_6H_4SMe)$ with blue light selectively yields electrically neutral versions of **trans-L2** and **cis-L2**. The single crystal X-ray diffraction (SCXRD) structures of the air-stable monocations of **trans-L2**, **cis-L2**, and **L6**, obtained via oxidation using ferrocenium, are shown in Figure 5.1a–c. Despite differing surface ligands, these clusters have nearly identical physical and electronic properties. Crystal structures indicate that the cluster cores remain structurally similar with negligible differences in Co–Co and Co–Se bond lengths or the maximum S⋯S distances. Furthermore, cyclic voltammetry (CV) of each cluster show identical reversible redox processes at similar potentials.[167]

We formed single-cluster junctions of **trans-L2**, **cis-L2**, and **L6** with ~1 μM solutions in propylene carbonate using the STM-BJ technique as described previously.[23] Briefly, a gold STM tip is brought in contact with a gold substrate (~1 cm^2) and subsequently retracted. Upon tip retraction, the gold-gold contact undergoes necking and ultimately ruptures, yielding two nanoscopic electrodes separated by ~1 nm. A single cluster from the surrounding solution can bridge the two electrodes via the anchoring ligands, thereby establishing a single-cluster junction through which current is measured. Measurements are performed on thousands of single-cluster junctions by repeatedly smashing and withdrawing the STM tip from the substrate. In order to

suppress background ionic current of the solvent, we coat the tip with an insulating wax;[210] this leaves an exposed surface area of ~10 μm^2. As a consequence of the large difference in the surface areas of the tip and substrate electrodes, the voltage drop across the junction is asymmetric, and the redox levels of the cluster are effectively pinned to the chemical potential of the substrate. This allows the potential applied to the tip to dictate the charge state of the cluster.[32, 132]

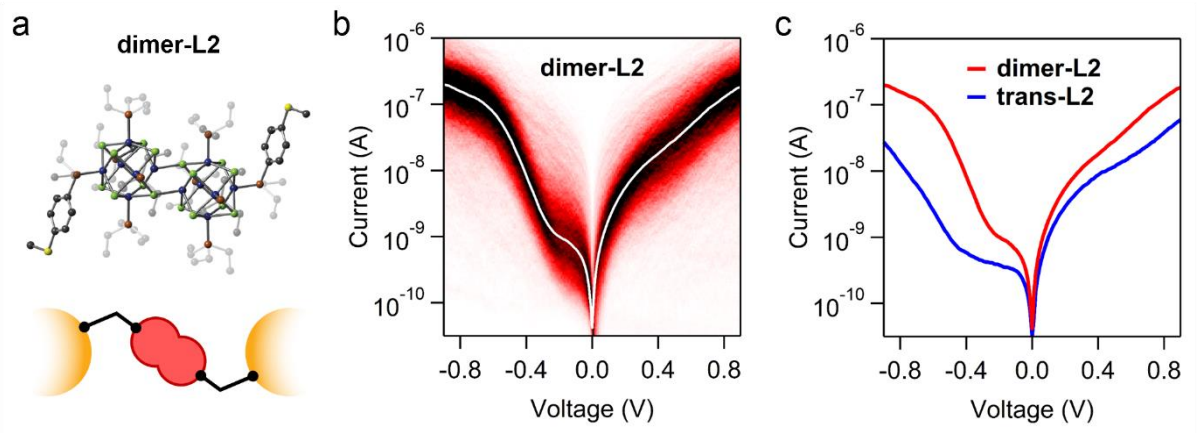

Figure 5.2. (a) Solid state structure of **dimer-L2** as determined from SCXRD as well as schematic molecular junction. (b) 2D absolute current versus voltage histogram for **dimer-L2** compiled from 4,355 traces. The current is logarithmically binned (40 bins/decade) and the voltage is linearly binned (100 bins/V). Average I-V curve is overlaid in white. (c) Average I-V curves representing most probable current versus voltage for **trans-L2** (blue) and **dimer-L2** (red).

To obtain the I-V characteristics, a single cluster was held between the tip and substrate at a constant distance and the current was recorded while the voltage was swept between +1 V and –1 V at a rate of 40 mV/ms. Several thousand I-V traces were collected for **trans-L2** (Figure 5.1d), **cis-L2** (Figure 5.1e), and **L6** (Figure 5.1f). All three clusters yield similar nonlinear I-V characteristics with a distinguishable current plateau feature spanning from 0.0 to –0.4 V. These results represent a significant deviation from traditional single-molecule devices that conduct through a coherent tunneling mechanism. When tunneling across a single-molecule junction is coherent, changing the substitution pattern, e.g., meta- vs para-substituted benzene, in general yields different electronic characteristics.[11] The similarity in the I-V characteristics for **trans-L2**

and **cis-L2** (Figure 5.1d,e) provides strong evidence for an incoherent transport mechanism, that is, a mechanism in which the electron relaxes on the cluster as it transits the junction. Additionally, the I-V characteristics of **L6** are comparable to those of **trans-L2** and **cis-L2** (Figure 5.1f).[204] These data suggest that despite there being several binding modes available to the system, the charge transport mechanism is unchanged. These results demonstrate that the Co_6Se_8 cores of these clusters function as ideal nanoscale circuit elements; they display identical electronic properties regardless of how they are wired in their circuits.

Next, we investigate the impact cluster size has on the I-V characteristics. Enabled by recent developments,[211] we synthesized a larger cluster, [*trans*-$Co_{12}Se_{16}(PEt_2(p\text{-}C_6H_4SMe))_2(PEt_3)_8][PF_6]$ (**dimer-L2**). The core of **dimer-L2** is a fused dimer of Co_6Se_8 octahedrons with two Co–Se intercluster bonds where a Co atom of one cluster is bound directly to a Se atom of another cluster (Figure 5.2a). In contrast to analogous structures with different metals,[212] the Co–Se intercluster bonds are shorter than the Co–Se intracluster bonds. As such, the individual Co_6Se_8 units are electronically linked and behave as a larger, cohesive cluster with a greater number of redox states than the monomeric clusters. CV demonstrates that the core of **dimer-L2** can assume oxidation states spanning 1^- to 4^+,[211] whereas all monomeric species can only assume the neutral to 3^+ oxidation states.[167]

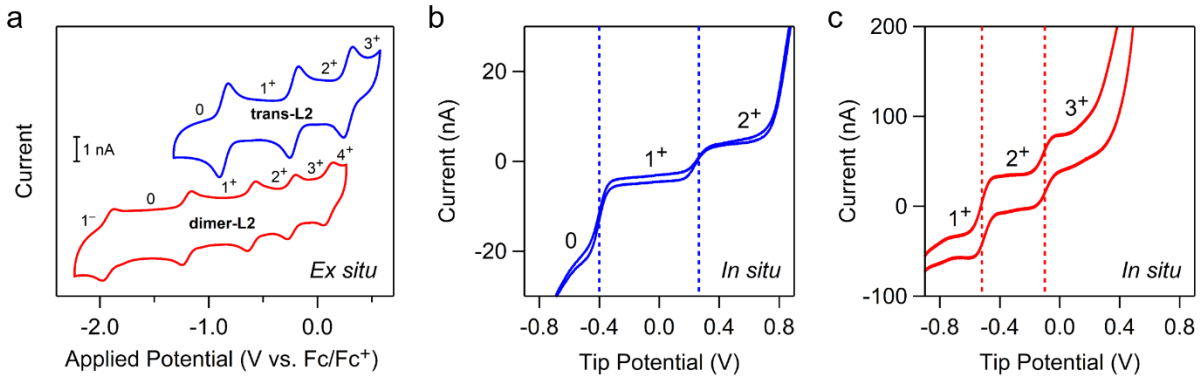

Figure 5.3. (a) *Ex situ* CV for **trans-L2** (blue) and **dimer-L2** (red). (b,c) *In situ* CV recorded for a

~0.1 mM solution of (b) **trans-L2** and (c) **dimer-L2** in PC. Current flowing into the wax-coated working electrode STM tip was measured while a linear voltage sweep (100 mV/s) was applied to the substrate. Vertical dashed lines highlight the positions of the redox transitions.

The positions of the junction-anchoring ligands on **dimer-L2** make for suitable comparison with **trans-L2**. We present a 2D histogram compiled from thousands of I-V measurements of **dimer-L2** in Figure 5.2b. As with **trans-L2**, the I-V characteristics for **dimer-L2** are highly nonlinear with a current plateau feature at low negative tip bias. However, the current plateau feature spans a smaller bias range from 0.0 to –0.2 V (versus 0.0 to –0.4 V in **trans-L2**). We directly compare the average I-V curves for **trans-L2** and **dimer-L2** in Figure 5.2c. In addition to the smaller current plateau feature, **dimer-L2** exhibits higher currents at all biases. The same relationship is observed in standard conductance measurements, where the conductance of **dimer-L2** is higher than that of **trans-L2**. This observation is in sharp contrast to the typical off-resonant coherent tunneling mechanism where adding repeat units to the conducting path results in a decreased conductance.[11] Furthermore, at high negative tip bias, the current saturates in **dimer-L2**, and a second plateau emerges in the I-V curve. Both observations can be explained by a sequential incoherent tunneling mechanism and the electronic and structural properties of the larger **dimer-L2**. As detailed later, the dimeric cluster features smaller reorganization energies upon charging/discharging, which results in a higher current.

To explore the relationship between the transport properties and the redox properties, we performed *in situ* CV measurements using the STM-BJ setup for both **trans-L2** and **dimer-L2**. The measurements were carried out with ~0.1 mM solutions of the clusters in propylene carbonate with the wax-coated gold tip withdrawn from the gold substrate. This setup functions as a two electrode electrochemical cell where the tip acts as a micro-scale working electrode and the substrate acts as the counter electrode.[213] Both clusters exhibit two reversible redox transitions

within the bias window relevant to the I-V measurements. The *in situ* CV is consistent with the *ex situ* CV, and the redox states can be assigned (Figure 5.3). Note that since the *in situ* CV does not have a reference electrode, only the relative redox potentials should be considered as the absolute potentials may shift. Importantly, the difference between the redox transitions agrees well with the bias range of the current plateau observed for both **trans-L2** and **dimer-L2**. At low bias, within the current plateau region, **trans-L2** is in the 1^+ oxidation state and **dimer-L2** is in the 2^+ oxidation state. As the bias is increased outside of the plateau region, oxidation or reduction of the clusters is enabled and a nonlinear increase in current occurs. Since the electrochemical window for the 2^+ oxidation state of **dimer-L2** ($\Delta E \sim 0.42$ eV) is smaller than for the 1^+ oxidation state of **trans-L2** ($\Delta E \sim 0.66$ eV), the I-V for **dimer-L2** displays a shorter current suppression plateau.

The charge transport behavior of these clusters can be described by a well-established model for hopping transport[205, 214-215] derived from heterogeneous electron transfer theory.[216-217] For the semi-classical Marcus model,[218-220] the rate of electron transfer for an electron of energy E between a metal and molecule for redox level i is characterized by a Gaussian function,[217]

$$D_i^{\pm}(E) = \frac{1}{\sqrt{4\pi\lambda_i k_B T}} \mathrm{Exp}\left[-\frac{\left(E - (E_i \pm \lambda_i)\right)^2}{4\lambda_i k_B T}\right] \tag{5.1}$$

where E_i is energy of the redox transition and λ_i is the reorganization energy. $(E_i + \lambda_i)$ denotes adding an electron to the molecule (reduction), and $(E_i - \lambda_i)$ denotes removing an electron from the molecule (oxidation).[3, 221-223]

Sequential tunneling across a molecular junction can be modeled by a set of rates of hopping on and off the cluster from both the left (L) and right (R) electrodes (Figure 5.4a,b). The rate of hopping on the cluster from one of the electrodes (reduction) is given by[205, 224-226]

$$k_{i,L(R)}^{+} = \frac{2\pi}{\hbar} \int_{-\infty}^{\infty} |t(E)|^2 \rho_{L(R)}(E) f_{L(R)}(E) D_i^{+}(E)\, dE \tag{5.2}$$

where $f_{L(R)}(E) = \left[1 + e^{(E-\mu_{L(R)})/k_B T}\right]^{-1}$ is the Fermi-Dirac distribution, $t(E)$ is the tunneling matrix element, and $\rho_{L(R)}(E)$ and $\mu_{L(R)}$ are the density of states and chemical potential of the electrodes, respectively. Similarly, the rate of hopping off the cluster to one of the electrodes (oxidation) is given by

$$k_{i,L(R)}^{-} = \frac{2\pi}{\hbar} \int_{-\infty}^{\infty} |t(E)|^2 \rho_{L(R)}(E) \left[1 - f_{L(R)}(E)\right] D_i^{-}(E)\, dE \tag{5.3}$$

We modeled the I-V characteristics of **trans-L2** and **dimer-L2** using two redox levels ($i = 1, 2$) in accordance with the two redox transitions observed in each *in situ* CV (Figure 5.4c,d). The hopping rates for the two-level model (three charge states) can be incorporated into a master equation which can be used to calculate the steady-state current as a function of applied voltage, given by[205, 227-228]

$$I(V) = e\left(k_{2,L}^{-} P(2) - k_{2,L}^{+} P(1) + k_{1,L}^{-} P(1) - k_{1,L}^{+} P(0)\right) \tag{5.4}$$

where $P(n)$ is the ensemble distribution of the number of electrons (n) occupying the model cluster and e is the charge of the electron.[229]

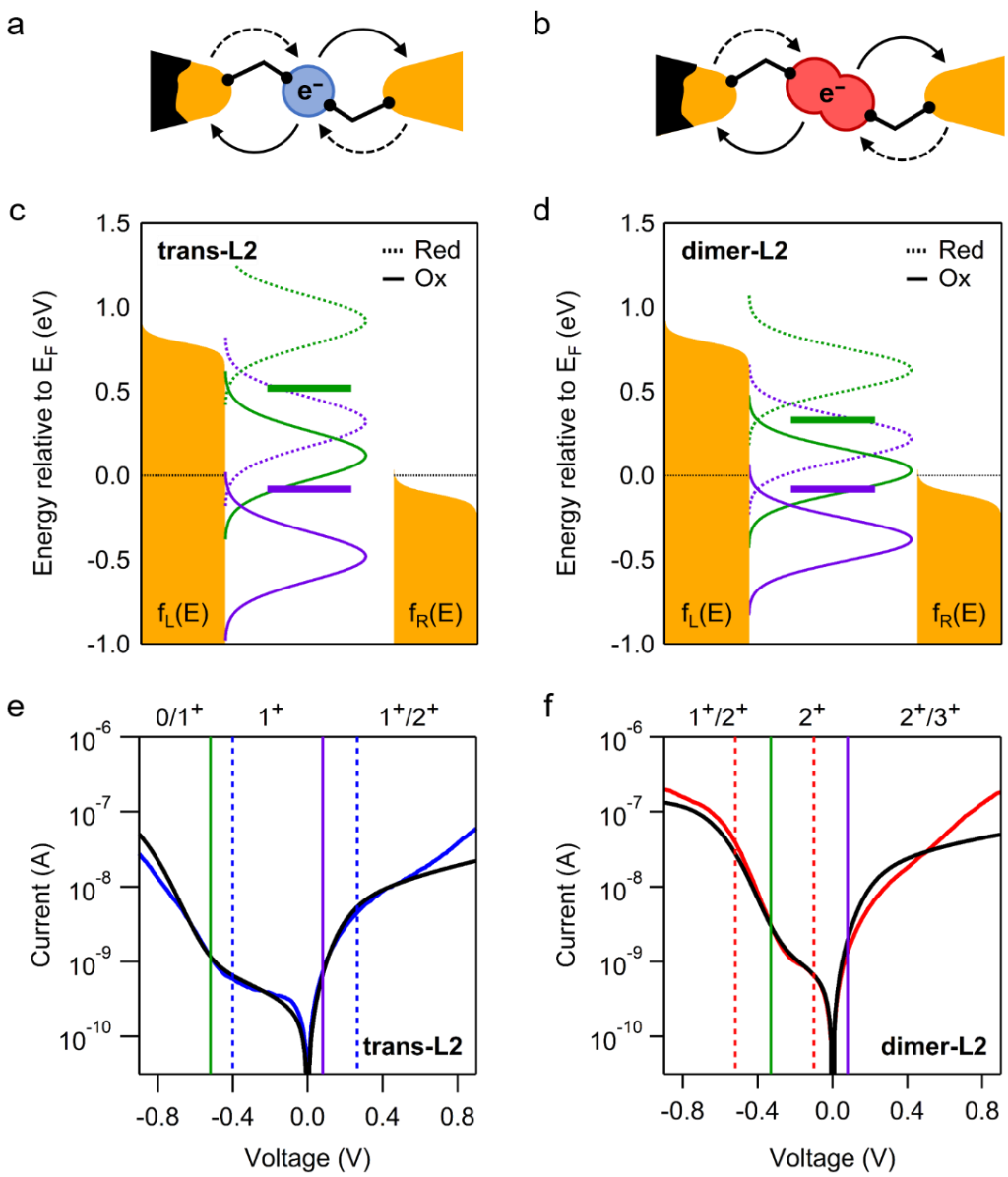

Figure 5.4. (a,b) Schematic of (a) **trans-L2** and (b) **dimer-L2** molecular junctions. The solid and dashed arrows depict the oxidation and reduction processes, respectively. The left electrode is the coated tip and right electrode is the substrate. (c,d) Schematic of the two-level model (i = 1, purple; i = 2, green) used to fit the I-V of (c) **trans-L2** and (d) **dimer-L2**. The Gaussian functions $D_i^{\pm}(E)$ are included in the schematic as well. The width of the Gaussians for **dimer-L2** are narrower due to the smaller reorganization energy. (e,f) Comparison of model fit (black) to experimental I-V curve for (e) **trans-L2** (blue) and (f) **dimer-L2** (red), with the oxidation state(s) for the cluster indicated. The vertical solid lines represent the values of E_1 (purple) and E_2 (green) from the model fit, and the vertical dashed lines represent the experimental redox transitions from Figure 5.3b,c. The spacing of the redox transitions are in good agreement between the model fit and experiment.

The fits to the experimental I-V curves are presented for **trans-L2** (Figure 5.4e) and **dimer-L2** (Figure 5.4f), showing excellent agreement to the experimental data. The positions of the two redox levels are demarcated by the solid vertical lines. These values relative to the Fermi energy (E_F) are: $E_1 = -0.08$ eV, $E_2 = 0.52$ eV for **trans-L2** and $E_1 = -0.08$ eV, $E_2 = 0.33$ eV for **dimer-L2**. For comparison, the positions of the redox levels from Figure 5.3b,c are demarcated by the dashed vertical lines. The spacing between the model levels for both **trans-L2** ($E_2 - E_1 = 0.60$ eV) and **dimer-L2** ($E_2 - E_1 = 0.41$ eV) are consistent with the *in situ* CV data ($\Delta E = 0.66$ eV for **trans-L2** and $\Delta E = 0.42$ eV for **dimer-L2**). Note that shifts in the redox potentials from *in situ* CV are reasonable due to the lack of a reference electrode. Information about the absolute redox potentials comes from the *ex situ* CV, which indicates nearly identical oxidation potentials for [**trans-L2**]$^{1+}$ and [**dimer-L2**]$^{2+}$ (Figure 5.3a). Therefore, the use of $E_1 = -0.08$ eV for both **trans-L2** and **dimer-L2** is justified. We note that the model deviates from the experimental data at large positive tip biases. This may be due contributions to the current from higher oxidative redox transitions that are unaccounted for in this simplified two redox level model.

The values used for the reorganization energies are also within what is expected for molecular clusters:[230-231] $\lambda_{1(2)} = 0.4\ eV$ for **trans-L2** and $\lambda_{1(2)} = 0.3\ eV$ for **dimer-L2**. The smaller reorganization energy for the larger cluster **dimer-L2** is consistent with the notion that the cluster with more atoms should undergo smaller structural distortions as a response to a change in redox state. *Ex situ*, the reorganization energy can be shown indirectly from the changes in average Co–Co bond lengths upon oxidation or reduction. In the Co_6Se_8 core of **trans-L2**, for every oxidation of the cluster, an average contraction of ~0.04 Å is observed from crystal structure analysis.[204] By contrast, the **dimer-L2** core shows no discernable change in cluster bond lengths over these oxidation states, fully consistent with the results from the fits to the model.[211] The

modeling demonstrates that the increase in current for **dimer-L2** can be ascribed to the smaller reorganization energy, which is in agreement with previous reports.[232-234]

In addition to higher currents, **dimer-L2** exhibits current saturation at high negative bias (Figure 5.2b). This current saturation is also reproduced by the model (Figure 5.4f). For the model, the maximum current that can be driven through the junction is given by,[205]

$$I^{sat} = \frac{2\pi e}{\hbar}\rho(E_F)|t|^2 \tag{5.5}$$

where we have assumed that $t(E) \approx t$ and that $\rho_{L(R)}(E) \approx \rho(E_F)$, which is reasonable for gold electrodes.[226] The current saturation I^{sat} is a manifestation of the inverted Marcus regime in the context of heterogeneous electron transfer, whereby excessively energetic electrons hop onto the cluster at an increasingly slower rate.[187, 213] Since **trans-L2** and **dimer-L2** utilize the same anchoring ligand, i.e., the same t, the same current saturation is expected for both molecular junctions. A value of $I^{sat} = 3 \times 10^{-7} A$ is used to model both **trans-L2** and **dimer-L2**, as this value fits the data best. For **trans-L2**, the current saturation is not observable within the experimentally accessible bias window. However, for **dimer-L2**, due to its smaller reorganization energy (λ_2) and smaller reduction potential (E_2), we are able observe a plateau in the current at large negative biases. Accordingly, **dimer-L2** exhibits a higher on/off ratio than **trans-L2**. Comparing the current at −0.15 V ("off") and −0.9 V ("on"), the on/off ratios are 78 and 230 for **trans-L2** and **dimer-L2**, respectively.

To validate I^{sat}, we can independently assess its value by estimating the tunneling matrix element (t). An approximate relationship between the low bias conductance of the anchoring ligand and the tunneling matrix element is shown in Eq. 5.6, where $G_0 = 2e^2/h$ is the quantum of conductance.[235]

$$G_{ligand} \approx 4\pi^2 \rho_L(E_F)\rho_R(E_F)|t|^2 G_0 \tag{5.6}$$

The conductance of the anchoring ligand (G_{ligand}), has been directly measured using the STM-BJ technique: $G \sim 5 \times 10^{-3} \mathrm{G_0}$.[208] Using $\rho_{L(R)}(E_F) = 0.3\ \mathrm{eV}^{-1}$ for gold[226, 236] yields a value for the tunneling matrix element of $t = 37.4$ meV, which corresponds to a saturation current $I^{sat} = 6.4 \times 10^{-7}$A using Eq. 5.5. This estimate is in good agreement with the saturation current observed experimentally for **dimer-L2** as well as the value used in the models ($I^{sat} = 3 \times 10^{-7}$A).

Lastly, the I-V characteristics for **trans-L2** and **dimer-L2** are not only highly nonlinear but also asymmetric, with rectification ratios of 10 for **trans-L2** and 6 for **dimer-L2** at ±250 mV. Symmetry between the forward and reverse currents is broken by the coated tip which establishes an asymmetric ionic environment in a polar solvent.[237] We can model the influence of the coated tip by means of a voltage division factor, $0 < \eta < 1$, that describes the asymmetric voltage drop resulting from the asymmetric ionic environment.[205, 238-239] The voltage division factor (or lever arm[187]) is incorporated into the chemical potentials of the left and right leads as $\mu_L = E_F - \eta eV$ and $\mu_R = E_F + (1 - \eta)eV$, respectively. When $\eta = 0$, the voltage drops at the right electrode, whereas when $\eta = 1$, the voltage drops at the left electrode. Here, the left electrode represents the coated tip (Figure 5.4a,b), and a value of $\eta = 0.88$ is found to fit the data best. This agrees with the assumption that, due to the small surface area of the tip electrode, most of the voltage drops between tip and the molecule.[32] In Figure 5.4c,d, the chemical potentials of the tip (μ_L) and substrate (μ_R) are depicted for a 0.9 V bias with $\eta = 0.88$. We note that the coated tip alone does not yield current rectification. The current rectification observed for the clusters measured herein results from the asymmetric positions of the redox levels relative to E_F. In contrast, no current rectification was observed for previously measured Co_6S_8 clusters, for which the redox levels were equally separated from E_F.[204]

We have studied the charge transport properties of atomically-precise inorganic clusters with varying surface ligand structure and core composition. The identical transport behaviors of **trans-L2**, **cis-L2**, and **L6**, as well as agreement with a model drawn from Marcus theory, firmly establish a sequential tunneling mechanism. Current is driven in these devices via an electron tunneling from the source electrode to the cluster through one anchoring ligand, and then a sequential tunneling step to the drain electrode via another anchoring ligand. Increasing the cluster core size from **trans-L2** to **dimer-L2** yields higher currents due to a reduction in both the reorganization energies and the energetic separation of the charge states. Additionally, **dimer-L2** exhibits current saturation at large negative bias, which further substantiates the proposed Marcus-based transport mechanism.

The single cluster devices exhibit practically-relevant properties including nonlinear I-V characteristics and current rectification, and our work establishes design principles through which these device properties can be engineered. By modifying the anchoring ligand or cluster composition through synthetic chemistry, one can tune the position and width of the current suppression plateau, the nonlinearity of the I-V, as well as the extent of current rectification. Moreover, the single-electron character of the devices opens up the possibility of performing room temperature single-electron manipulations in a molecular junction. Lastly, by increasing the cluster size beyond **dimer-L2**, the transition from discrete molecule to extended material can be explored.

Chapter 6: Photoconductivity in single-molecule junctions

6.1 Preface

This chapter discusses a novel experimental setup that combines the scanning tunneling microscope-based break junction technique (STM-BJ) with surface plasmon resonance imaging (SPRi). I would like to thank Professor Milan Delor for valuable discussions, and I would like to acknowledge Professor James Hone and Professor Julio Fernandez for providing optical equipment. Figure 6.2 was adapted from an optical diagram created by Woojung Lee.

6.2 Introduction

The conductance of a single-molecule junction can be modified by the absorption of photons, producing a photoconductivity effect.[13, 240] Observing photoconductivity is of key scientific interest as it would further elucidate the mechanisms of electron transport in single-molecule junctions.[241-242] Photoconductivity is believed to arise from three mechanisms. First, photon absorption can occur within the molecule and induce electronic transitions, which can in principle lead to phenomena such as resonant current amplification.[13, 243] Second, photon absorption can occur in the metal electrodes, generating so-called hot electrons/holes.[244-245] These energetic charge carriers can tunnel across the junction with an altered probability, thereby modifying the measured conductance. The third mechanism is known as photon-assisted tunneling (PAT).[246-247] In PAT, the light is viewed as an AC potential that adiabatically modulates the energy levels in the electrodes. For each energy level, the AC potential generates a series of virtual energy levels that are separated by $\hbar\omega$, where ω is the angular frequency of the light. The conductance is modified by the presence of these virtual energy levels, similar to the mechanism for hot electron/holes. Although all three photoconductivity mechanisms have been observed,[248-251] rigorous demonstrations of each mechanism, and the interplay among them, remains elusive.

Herein, we present a novel scanning tunneling microscope break-junction (STM-BJ) platform that will enable comprehensive investigations of photoconductivity in single-molecule junctions.

The central challenge in photoconductivity experiments is the relatively weak interaction between free-space light and the molecular junction. For example, the absorption cross-section for a single-molecule at room temperature is rather small,[252] and hot carrier generation in metals is often inefficient.[253] However, by coupling to surface plasmon-polaritons (SPPs), light can be strongly concentrated to the vicinity of the molecular junction, greatly enhancing any light-matter interactions.[254] For this reason, it is believed that SPP excitation is key to observing a measurable photoconductivity in single-molecule junctions.[13, 255-256]

SPPs are coupled oscillations between light and surface charges on a metal.[257] These oscillations, once excited, can propagate along the dielectric/metal interface. The discussion herein will focus on SPPs propagating along the air/gold interface, which is the interface relevant to the STM-BJ measurement (without solvent). Exciting SPPs is challenging due to the wavenumber mismatch between light and the SPP with the same frequency. The wavenumber mismatch between light propagating in air and the SPP at the air/gold interface is quantified by their dispersion relations (Figure 6.1a). For a given angular frequency ω, the wavenumber of light in air (k_{air}) is given by Eq. 6.1, where $n_{air} \approx 1$ is the refractive index of the air, and c is the speed of light. The corresponding wavenumber of the SPP ($k_{SPP}^{air/gold}$) is given by Eq. 6.2, where $\varepsilon_{air} \approx 1$ is the relative permittivity of the air, and $\varepsilon_{gold}(\omega)$ is the relative permittivity of gold. Classically, the gold can be modeled as $\varepsilon_{gold}(\omega) = 1 - \omega_p^2/\omega^2$, with plasma frequency ω_p ($\hbar\omega_p \approx 9\ eV$ for gold).[258] Eq. 6.2 demonstrates that $k_{SPP}^{air/gold}$ is necessarily greater than k_{air}; or equivalently, the wavelength of the SPP is smaller than normal light (Figure 6.1a).

$$k_{air} = n_{air}\left(\frac{\omega}{c}\right) \tag{6.1}$$

$$k_{SPP}^{air/gold} = k_{air}\sqrt{\frac{\varepsilon_{air}\varepsilon_{gold}(\omega)}{\varepsilon_{air} + \varepsilon_{gold}(\omega)}} > k_{air} \tag{6.2}$$

The wavenumber mismatch can be overcome through the use of a grating; however, installing a grating, on either the tip or substrate, requires time-consuming lithography.[249, 259] A more convenient approach is to couple to SPPs using a prism that has a higher refractive index than air ($n_{prism} > n_{air}$). For the prism-coupling scheme, a thin film of gold (~ 45 nm) is deposited on a prism, and SPPs are excited on the air/gold interface via p-polarized laser irradiation through the prism.[260] The higher refractive index of the prism increases the wavenumber of the light ($k_{prism} = n_{prism}(\omega/c)$) relative to that in air (Eq. 6.1), enabling SPP excitation provided that $k_{prism} > k_{SPP}^{air/gold}$.

$$k_{prism}^{x}(\theta) = k_{prism}\sin\theta \tag{6.3}$$

$$k_{air} = k_{prism}^{x}(\theta_{Crit}) < k_{SPP}^{air/gold} < k_{prism}^{x}(\theta_{NA}) < k_{prism} \tag{6.4}$$

To exactly match $k_{SPP}^{air/gold}$, the in-plane wavenumber of the light in the prism (k_{prism}^{x}) can be tuned by adjusting the angle of incidence (θ), as described by Eq. 6.3. The prism-coupling scheme relies on the phenomena of total internal reflection, which occurs above a critical angle. The critical angle (θ_{Crit}) is provided by the formula: $\sin\theta_{Crit} = n_{air}/n_{prism}$. At the critical angle, k_{prism}^{x} (Eq. 6.3) is equivalent to the maximum in-plane wavenumber in air ($k_{prism}^{x}(\theta_{Crit}) = k_{air}$), which is insufficient to excite SPPs (Eq. 6.4). Therefore, in the prism-coupling scheme, SPP excitation occurs above the critical angle. Above the critical angle, laser irradiation produces an evanescent electric field (normal to the surface) that can penetrate the gold film (Figure 6.1b). At the appropriate incident angle ($k_{prism}^{x}(\theta_{SPR}) = k_{SPP}^{air/gold}$), SPPs can be excited with near unit efficiency and the reflectance of the gold film approaches zero.[257] This phenomenon is known as

surface plasmon resonance (SPR). Prism-coupling provides a very precise and efficient scheme for exciting SPPs on thin gold films.

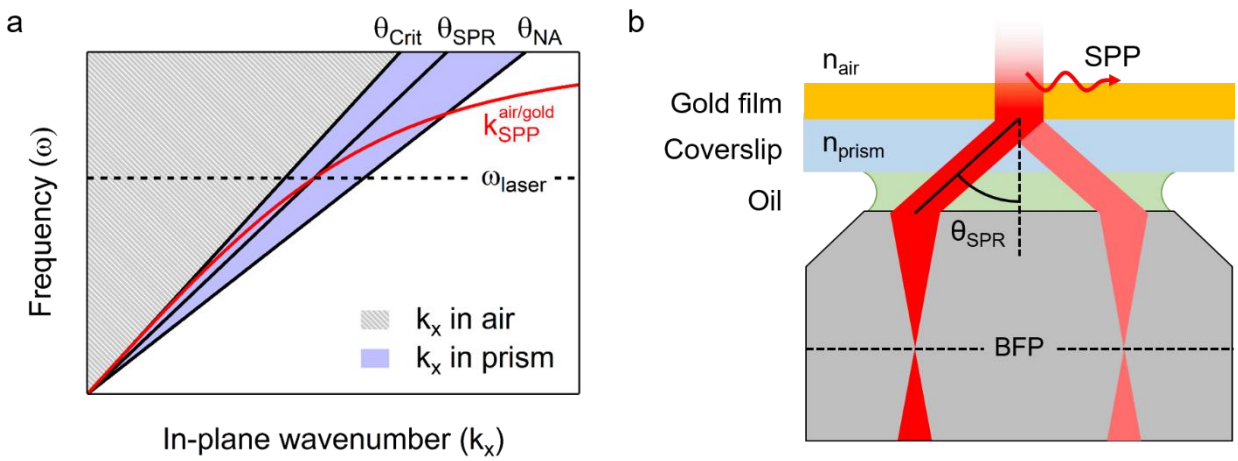

Figure 6.1. a) Dispersion relation for the SPP (red) compared to light in different media. The frequency of interest ω_{laser} is the frequency supplied by the laser. The in-plane wavenumber for light in air ($k_{air}^{x}(\theta) = k_{air} \sin\theta$) is too small regardless of incidence angle θ (gray shaded region). Using a prism (i.e., an oil immersion objective), larger wavenumbers ($k_{prism}^{x}(\theta)$) can be accessed (blue shaded region) if θ is greater than the critical angle (θ_{Crit}). The maximum incidence angle (θ_{NA}) is dictated by the numerical aperture (NA) of the objective. SPPs can be excited via the prism-coupling scheme at the SPR angle (θ_{SPR}), where $k_{prism}^{x}(\theta_{SPR}) = k_{SPP}^{air/gold}$. b) Schematic of the prism-coupling scheme using an oil-immersion objective (gray). The laser (red) is displaced from the optical axis and is focused onto the back focal plane (BFP) of the objective. The immersion oil (green) and the glass coverslip (blue) have the same refractive index (n_{prism}).

To interface the prism-coupling scheme with the STM-BJ technique, we employ an inverted microscope setup that can perform surface plasmon resonance imaging (SPRi). In an SPRi setup, the prism takes the form of an oil-immersion lens, and the laser is focused to the back focal plane of the objective, allowing the gold film to be irradiated by a collimated beam (Figure 6.1b).[261] The angle of incidence can be adjusted by moving the laser source using a translation stage. SPP excitation can be identified using a camera that captures the reflected light. At the SPR angle, the reflectance decreases substantially, and the camera image is dark. The upper bound on the angle of incidence (θ_{NA}) is set by the numerical aperture (NA) of the objective. The numerical aperture

is defined by the equation: $NA = n_{prism} \sin(\theta_{NA})$. Therefore, the frequency of the laser (ω_{laser}) must be sufficiently low so that $k_{SPP}^{air/gold}$ is smaller than $k_{prism}^{x}(\theta_{NA})$ (Eq. 6.4). Typically, red or near-infrared lasers are used to excite surface plasmons.

STM-BJ can be interfaced with SPRi by mounting an STM head on top of the SPRi setup, with the gold-coated coverslip serving as the substrate electrode. The STM tip can be positioned in the laser spot, and break-junction measurements can be performed in the presence of SPPs. The hybrid STM-BJ/SPRi setup provides a convenient method to investigate photoconductivity effects arising from SPP excitation. Herein, the details of the STM-BJ/SPRi setup will be discussed as well as some preliminary results demonstrating SPP excitation.

6.3 Materials and methods

STM-BJ measurements require significant isolation from mechanical and electrical noise. Several layers of noise isolation are implemented in the STM-BJ/SPRi setup. The setup is mounted on a passive vibration isolation table (Technical Manufacturing Corporation) and is enclosed in a home-built box. The box consists of aluminum sheets that are supported by a T-slot aluminum frame (80/20 Inc.). The metal casing serves as a Faraday cage that shields the setup from electrical noise. The inside of the box is covered with sound-absorbing foam (McMaster-Carr) to dampen any acoustic noise. The STM-BJ/SPRi setup is mounted on an optical breadboard. The optical breadboard is placed on a table-top active vibration isolation table (Table Stable Ltd.) to further dampen any high frequency mechanical noise. Additionally, an inner tube is sandwiched in between the optical breadboard and the table-top isolation table to dampen any low frequency mechanical noise. Together, these measures provide sufficient isolation from mechanical and electrical noise to perform STM-BJ measurements.

6.3.1 Optical setup

A detailed schematic of the STM-BJ/SPRi setup is provided in Figure 6.2. The STM tip is mounted on an XYZ translation stage (Newport Corporation). The gold-coated coverslip is mounted on an XY translation stage using microscope clips. The microscope clips provide both mechanical and electrical contact. The coverslip is held at a fixed height, and the oil-immersion objective can be brought into focus using a Z translation stage. The oil-immersion objective (Zeiss, 63X, 1.4 NA) is mounted in a lens turret along with lower magnification lenses (i.e., 10X, 20X, and 40X). The lower magnification lenses aid in the alignment of the tip to the optical axis.

The optical systems for the laser and the camera are separated using a pellicle beam splitter. The pellicle beam splitter consists of a thin nitrocellulose membrane. The thin membrane reduces any optical aberrations that would accompany a thicker beam splitter cube. The image of front focal plane of the objective (i.e., the gold film) is projected onto the camera using a tube lens (148 mm working distance). Conversely, the back focal plane can be imaged by placing an additional lens (a so-called Bertrand lens) in between the tube lens and the camera. This Bertrand lens is mounted on a 90° flip mount, enabling convenient toggling between front and back focal plane imaging (Figure 6.2).

A diode laser (Thorlabs Inc.) is used as the laser source. The diode laser is mounted in a thermoelectric cooler (Arroyo Instruments) that maintains the diode temperature at 25 °C. The laser is mounted on an XYZ translation stage that is additionally mounted on a 3-axis tilt platform. This enables the orientation of laser to be adjusted in all six degrees of freedom. The 3-axis tilt platform aligns the laser to the optical axis of the microscope objective. The XYZ translation stage allows the angle of incidence of the laser to be adjusted. The laser is passed through a polarizer for greater experimental control. Additionally, a pinhole is employed to clean up the beam, which

improves imaging quality. The laser is focused onto the back focal plane of the objective using a pair of lenses whose positions can be easily adjusted (Figure 6.2).

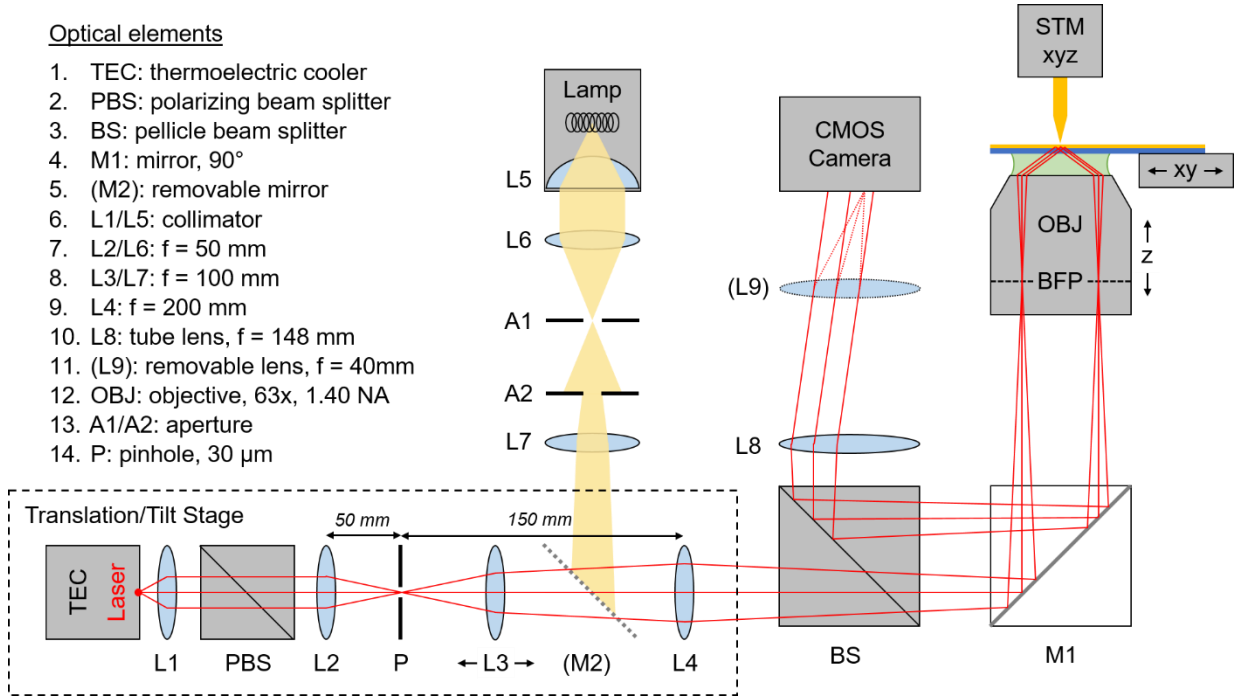

Figure 6.2. Optical diagram of the STM-BJ/SPRi setup. Using the translation/tilt stage, the laser is offset from the optical axis of OBJ. Using L3 and L4, the laser (660 nm) is focused on to the back focal plane (BFP). The laser can be focused any distance between 20 mm and infinity from L4 by adjusting the position of L3. The Bertrand lens (L9) can be added or removed to toggle between the BFP and the front focal plane.

Effective illumination from a white light source is essential for convenient alignment of the STM tip. Towards this end, we set up a Köhler illumination scheme, where the image of the halogen lamp is focused onto the back focal plane of the objective. Köhler illumination affords a uniform illumination of the sample, which allows one to resolve subtle features in the image such as the position of the STM tip. Instead of a beam splitter, a removable mirror is placed in the optical path of the laser when white light illumination is desired (Figure 6.2).

6.3.2 Substrate preparation

Uniform films of gold are necessary to observe the SPR effect. Evaporating uniform films of gold on glass is challenging because gold displays a poor wettability on untreated glass and tends to form discontinuous islands on the glass, which is undesirable for SPR. To improve gold adhesion to glass, an adhesion layer is often employed. Typically, a few nanometers of chromium or titanium is evaporated onto the glass prior to gold evaporation. Chromium and titanium can adhere to the glass through metal-oxide formation, and gold can subsequently alloy with these metals However, the presence of additional metals can complicate the plasmonic response of the gold film.[262] Instead, we employ a thiol monolayer as the adhesion layer between gold and glass.[263]

First, the coverslips (No. 1.5) are cleaned in a 1-2% solution of Hellmanex III detergent for at least six hours. The coverslips are then rubbed with a clean, gloved hand, rinsed with deionized water, and then dried with compressed air. These steps remove any grease and dust from the coverslips. Next, the dried coverslips are placed in a desiccator with an open vial of 3-mercaptopropyl trimethoxysilane (MPTMS). Vacuum is drawn on the desiccator for two minutes, and then the desiccation chamber is sealed and left at room temperature overnight. During this time, the MPTMS coats the coverslips through a vapor-phase silanization reaction. MPTMS silanization functionalizes the glass surface with thiols that can effectively adhere to gold.

The next day, prior to gold evaporation, the coverslips are baked at 85 °C for 10 min to remove any excess MPTMS, leaving only a monolayer. The thiol-functionalized coverslips are mounted in a home-built thermal evaporator, and the evaporation is performed at a base pressure 4×10^{-6} Torr. Importantly, the gold is deposited at a very fast deposition rate of ~ 10 Å/s. The goal is to flood the coverslip with gold to outcompete surface diffusion, which would cause undesirable island formation. Good quality gold films should have a golden color without a blue tinge. The

SPR resonance effect is optimized for a film thickness of ~ 45 nm,[257] but film thicknesses as low as 10 nm can be prepared with this procedure.[249, 263]

6.4 Results and discussion

As an initial demonstration, we image a gold-coated coverslip (48 nm gold film thickness). First, a collimated laser beam (660 nm) is passed to the oil-immersion objective, which focuses the laser to a small area on the gold film. The image of the back focal plane reveals a dark ring in the reflected light at the SPR angle (Figure 6.3a). Since the laser is polarized, the SPR effect is only visible in the p-polarized region and not in the s-polarized region. The dark ring has a narrow angular width, indicating that SPP excitation occurs for a very specific resonance condition.

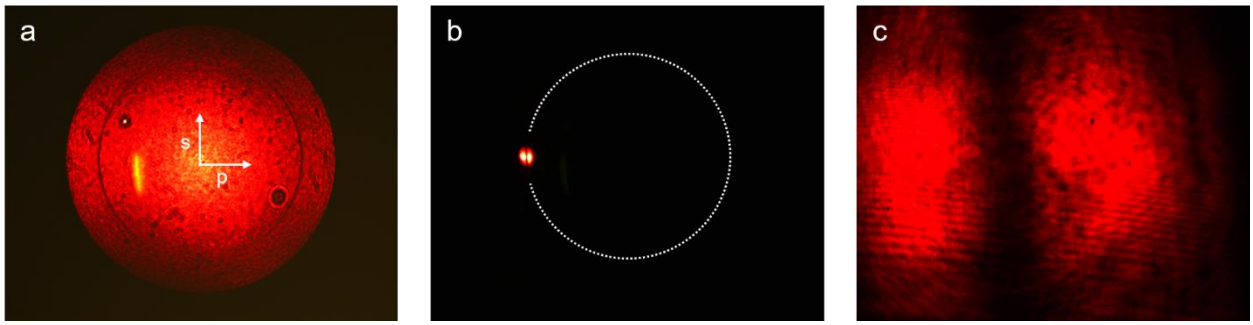

Figure 6.3. a) Back focal plane image of 660 nm laser focused on a 48 nm gold film. SPR appears as a dark ring in the image. The SPR is strongest in the p-polarized direction and weakest in the s-polarized direction. b) Back focal plane image of laser is focused on back focal plane at the SPR angle. The dark SPR ring in (a) is indicated by the dashed white line. c) Front focal plane image corresponding to (b). The dark band represents a region of SPP excitation.

To efficiently excite SPPs, the laser (660 nm) can be focused on to the back focal plane. This is achieved by simply adjusting the position of lens L3 (Figure 6.2). The laser source can then be translated until the laser spot on the back focal plane is positioned precisely at the SPR angle (Figure 6.3b). At this laser position, the front focal plane image (i.e., the gold film) exhibits a dark band (Figure 6.3c). The dark band represents a region where the laser is not reflected by the gold

film due to SPP excitation. Switching between the back focal plane (Figure 6.3b) and the front focal plane (Figure 6.3c) is achieved by flipping the lens L9 (Figure 6.2) open and closed.

The SPR angle is very sensitive to the refractive index of the dielectric (i.e., air). Typically, STM-BJ measurements are performed by adding molecule to the substrate, either dissolved in a solvent or as a dry film. However, coating the substrate with an organic material will shift the SPR angle outside of the numerical aperture of the oil-immersion objective. This would render the setup ineffective at exciting SPPs. To circumvent this issue, the STM tip is coated with a dry film of the molecule. During break-junction measurements, the refractive index of the air is only modified locally near the tip. To perform STM-BJ measurements in the presence of SPPs, the molecule-coated tip is positioned in the dark band of the laser spot (Figure 6.3c). STM-BJ measurements of molecules such as 4,4'-bipyridine exhibit a well-defined conductance peak. Unfortunately, as of yet, the conductance peak position does not depend on laser intensity.

In conclusion, we report the successful implementation of a hybrid STM-BJ/SPRi setup. STM-BJ measurements can be performed in the inverted microscope setup with negligible mechanical noise. Furthermore, SPPs can be efficiently excited using the oil-immersion objective, as evidenced by the dip in reflectance at the SPR angle. Unfortunately, initial measurements do not indicate a photoconductivity when the single-molecule junction is irradiated by an intense laser beam. Given the versatility of the setup, we remain hopeful that photoconductivity can be observed with the right molecule and the right illumination scheme.

Conclusions

In this dissertation, the mechanisms of electron transfer have been investigated in detail in single-molecule junctions. Experimental measurements demonstrated that the process of electron transfer depends strongly on the properties of the molecular bridge. Through accompanying theoretical modelling, it is shown that coherent, direct tunneling is operative for polymethine wires, whereas incoherent, sequential tunneling is operative for molecular clusters. These results confirm key principles of chemical physics in direct measurement of individual molecules.

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